

TWO-FOLD LANGFORD SEQUENCES EXIST EXACTLY WHEN $3l \geq 2d - 1$, AND WHAT IS KNOWN FOR EVERY MULTIPLICITY

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ABSTRACT. A two-fold Langford sequence of order l and defect d is a sequence of length $4l$ in which every $p \in [d, d + l - 1]$ occupies two disjoint pairs of positions at distance p . Alkasasbeh, Dyer and Howell asked for necessary and sufficient conditions for the existence of m -fold Langford sequences, $m \geq 2$. We answer the case $m = 2$: for $d, l \geq 1$ a two-fold Langford sequence of order l and defect d exists if and only if $3l \geq 2d - 1$. Necessity is a distance count. Sufficiency reduces, by concatenation and a mirror-symmetric ansatz, to a problem about permutations of $\{1, \dots, l\}$ whose signed displacements are prescribed; an inversion symmetry and a composition lemma reduce that problem to a band near its diagonal, which is closed by twenty-one block-reversal families and the two-block reversal τ_l , each proved by a class table, together with eleven explicit permutations. For every m we prove the counting bound $(2m - 1)l \geq 2d - 1$ with a parity condition for odd m , show that the bound is attained for every m and only by sequences in which every pair straddles the midpoint, and settle the order $l = 1$ (a sequence exists if and only if $d \mid m$). We prove a forced-endpoint bound and a family of necessary conditions, $r(T - r) \leq m \sum_{p=d}^{d+l-1} |p - T|$ with $r = ml \bmod T$ for every integer $T \geq 1$, which contains the counting bound and the necessity half of the order-one result. For every $m \geq 3$ the cell $(d, l) = (3m - 3, 3)$ satisfies counting and parity and has no sequence, so counting and parity are not sufficient. These statements are formalized and machine-checked in Lean 4. For $m \geq 3$ we also describe, with exact Farkas certificates and closed-form edges, the empty bands inside the counting bound, and give an exact finite test that detects every negative member of that certificate family; that part is proved on paper and not formalized, and the general characterization remains open.

1. INTRODUCTION

Write $[u, v]$ for the set of integers x with $u \leq x \leq v$. A *Langford sequence* of defect d and order l is a sequence of length $2l$ in which each $p \in [d, d + l - 1]$ occurs exactly twice, at two positions p apart; at $d = 1$ these are Skolem sequences [13, 11], and their existence was settled by Simpson [12]. Alkasasbeh, Dyer and Howell [1] introduced the *two-fold* version, in which every $p \in [d, d + l - 1]$ occupies two disjoint pairs of positions at distance p , so that the sequence has length $4l$; more generally an *m -fold* Langford sequence has length $2ml$ and m disjoint pairs for every p (the $d = 1$ case is the m -fold Skolem sequence of Baker, Nowakowski, Shalaby and Sharary [4]). Section 7 of [1] closes by asking for necessary and sufficient conditions for the existence of m -fold Langford sequences with $m \geq 2$. The only two-fold Langford family printed there (Construction 2.7, defect $6k - 1$, order $4k - 1$) lies on the extreme line of the counting bound.

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Theorem A. *Let $d, l \geq 1$. A two-fold Langford sequence of order l and defect d exists if and only if $3l \geq 2d - 1$.*

Theorem B. *Let $m, d, l \geq 1$.*

- (i) *If an m -fold Langford sequence of order l and defect d exists, then $(2m-1)l \geq 2d-1$, and if m is odd, $l(2d+l+1) \equiv 0 \pmod{4}$.*
- (ii) *If $(2m-1)l = 2d-1$, an m -fold Langford sequence of order l and defect d exists.*
- (iii) *An m -fold Langford sequence of order 1 and defect d exists if and only if $d \mid m$.*
- (iv) *No three-fold Langford sequence of order 3 and defect 6 exists.*
- (v) *For every $m \geq 3$ the conditions of (i) are not sufficient: some (d, l) satisfies them and has no m -fold Langford sequence.*
- (vi) *If an m -fold Langford sequence of order l and defect d exists, then for every integer $T \geq 1$, with $r = ml \bmod T$,*

$$r(T-r) \leq m \sum_{p=d}^{d+l-1} |p-T|.$$

- (vii) *If an m -fold Langford sequence of order l and defect d exists, then $6mld \leq 4d^2 + 2m^2l^2 + ml(l-1)$.*
- (viii) *For every $m \geq 3$ the cell $(d, l) = (3m-3, 3)$ satisfies the conditions of (i) and has no m -fold Langford sequence.*
- (ix) *If s is an m -fold Langford sequence of order l and defect d , then $(2m-1)l = 2d-1$ if and only if $ml < i + s_i$ for every $i \in [1, ml]$, that is, if and only if every pair has its left end in $[1, ml]$ and its right end in $[ml+1, 2ml]$.*

At $m = 2$ the parity clause of (i) is vacuous and Theorem A says that (i) is exact. At $m = 1$ the conditions of (i) are Simpson's, and exactness there is his theorem [12]; at $d = 1$ they are the conditions of [4]. Part (v) shows that nothing of the kind holds for $m \geq 3$, where counting and parity give no complete existence characterization; explicit witnesses are $(d, l) = (6, 3)$ at $m = 3$, $(m-1, 1)$ for even $m \geq 4$ and $(m-2, 1)$ for odd $m \geq 5$, and by (viii) they can be taken in order three for every m . Sufficient families do exist for every m : the tight line (ii) and the row $d \mid m$ of (iii). Part (vi) is a family of necessary conditions, one for every T , which contains the counting bound of (i) (take $T \geq \max(ml, d+l-1)$) and the necessity half of (iii) (take $l = 1$ and $T = d$). Part (vii) comes from the forced left ends $[1, d]$ and proves (viii) for $m \geq 4$, and (ix) says that the bound of (i) is attained only when every pair straddles the midpoint.

What is formalized. Theorem A and all nine parts of Theorem B are formalized and machine-checked in Lean 4 with Mathlib [7, 8], in the accompanying repository [15], as the theorems

<code>Langford.twoFold_exists_iff</code> (Theorem A)	<code>Langford.order_one_iff</code> (B(iii))
<code>Langford.necessary</code> (B(i))	<code>Langford.not_threeFold_six_three</code> (B(iv))
<code>Langford.tight_exists</code> (B(ii))	<code>Langford.not_sufficient</code> (B(v))
<code>Langford.residue_bound</code> (B(vi))	<code>Langford.forced_endpoint</code> (B(vii))
<code>Langford.not_order_three</code> (B(viii))	<code>Langford.tight_iff_straddle</code> (B(ix))

about the definition `Langford.IsLangford` (Definition 2.1). Every numbered statement below that carries a *Lean* line is proved there under the name given. Statements that carry a *Status* line instead are proved on paper in this note and are not formalized, or are finite computations; they are Remark 3.2, Corollary 3.5, Remark 3.8, Corollary 3.10, Lemma 4.6,

Lemma 7.5, the computations for $m \geq 3$ in §7, and the whole of §8. The note is the human-readable account of the formalized proof: it follows the route the formalization takes, so that a reader can check one against the other without reading Lean.

The shape of the proof of Theorem A. Necessity is the distance sum (Theorem 3.1): the right ends minus the left ends of the $2l$ pairs sum to $2 \sum p$, and no partition of $[1, 4l]$ into pairs does better than $4l^2$. For sufficiency put

$$e = 3l - 2d + 1 \text{ (the excess)}, \quad \delta = d - l, \quad \mu = 2\delta - 1 = l - e.$$

- Two sequences (d, l_1) and $(d + l_1, l_2)$ placed one after the other form $(d, l_1 + l_2)$ (Lemma 4.1); every in-bound cell with $l \geq 2d$ splits in this way into two smaller in-bound cells (Lemma 4.2).
- Every other in-bound cell with $l \geq 5$ has $-l \leq \mu \leq l$, and a mirror-symmetric ansatz (Lemma S, Lemma 4.5) turns a permutation σ of $[1, l]$ whose numbers $\sigma(w) - w + \delta$ and their mirrors tile $[1 - l, l]$ — the *signed-permutation problem* $\text{SP}(l, \delta)$ — into a two-fold sequence of defect $l + \delta = d$.
- Inversion $\sigma \mapsto \sigma^{-1}$ exchanges μ and $-\mu$; the reversal and a two-block reversal τ_l solve $\mu = 1$ and $\mu = l$; and a composition lemma (Lemma C) carries a solution at (n, μ) to one at $(n + \mu, \mu)$. So every cell descends along its line of fixed μ to a base cell in the *band* $l/2 < \mu \leq l$ (Theorem 4.11); one base cell, $(l, \mu) = (4, 3)$, is empty, and $(7, 3)$ is supplied instead.
- The band is closed, in each residue class of l modulo 4, by explicit block-reversal permutations whose block sizes are affine in (l, μ) : twenty-one families and τ_l , each proved by a *class table*, plus eight sporadic cells and two small bases (§5 and Appendix A).
- The sixteen in-bound cells with $l \leq 4$ are explicit.

No part of the argument depends on Simpson’s theorem.

Beyond $m = 2$. Section 3 proves Theorem B(i)–(iii), (vi), (vii) and (ix) and the juxtaposition lemma, which gives, with Theorem A, a sequence at every cell with $3l \geq 2d - 1$ for every even m (Corollary 3.5). Section 7 proves (iv), (v) and (viii), and §8 describes the picture for $m \geq 3$: an elementary emptiness lemma, a one-parameter family of integer Farkas certificates whose optimum yields closed-form empty bands with edges $d/l \rightarrow [(2j + 1)m - J \mp \sqrt{m^2 - J^2}]/(2J)$, $J = j(j + 1) < m$, a quadratic inequality that holds at every integer T without regime hypotheses, an exact finite test that detects every negative member of the family, realized cells between two empty bands, and a conjecture. Section 10 lists the prior art that was searched.

2. DEFINITIONS, FIDELITY AND CERTIFICATES

Definition 2.1. Let $m, d, l \geq 1$. A sequence $s = (s_1, \dots, s_{2ml})$ is an *m -fold Langford sequence of order l and defect d* if every entry lies in $[d, d + l - 1]$ and, for every $p \in [d, d + l - 1]$, there is a set A_p of m positions such that for every $a \in A_p$

$$1 \leq a, \quad a + p \leq 2ml, \quad a + p \notin A_p,$$

and the positions $i \in [1, 2ml]$ with $s_i = p$ are exactly the elements of A_p and of $A_p + p$. For $m = 2$ we say *two-fold*.

Lean: `Langford.IsLangford (Challenge.lean; verbatim copy in L2/Defs.lean).`

The formal definition reads s as a function $\mathbb{N} \rightarrow \mathbb{N}$ on the positions $1, \dots, 2ml$ and ignores its other values; intervals are written with bounds. The hypotheses $d \geq 1$ and $l \geq 1$ in Theorem A are needed: for $l = 0$ the empty sequence qualifies, and for $d = 0$ nothing does.

Lemma 2.2 (the pair form). *An m -fold Langford sequence of order l and defect d exists if and only if $[1, 2ml]$ can be partitioned into ml pairs $\{a, b\}$, $a < b$, in which every $p \in [d, d + l - 1]$ is the difference $b - a$ of exactly m pairs.*

Proof. Given s and the sets A_p , take the pairs $\{a, a + p\}$, $a \in A_p$. For fixed p these are m pairs with $2m$ distinct endpoints: the a are distinct, so are the $a + p$, and $a + p \notin A_p$ forbids a left end from being another pair's right end. For distinct p the endpoint sets are disjoint because a position carries one value. The l values give $2ml$ endpoints in $[1, 2ml]$, so the pairs partition it. Conversely, put s_i equal to the difference of the pair containing i and A_p equal to the set of left ends of the pairs of difference p . $\square$

Remark 2.3 (fidelity: chains). The definition of [1] asks, for each p , for “exactly 2 disjoint pairs” of occurrences at distance p , and in a second condition that the two occurrences in each pair be p apart. The second condition applies within the chosen pairs: read for all four occurrences it could never hold. The first is read as above: the occurrences of p *can be partitioned* into m pairs at distance p , so four occurrences at $a, a + p, a + 2p, a + 3p$ (a *chain*) are allowed. This is the m -fold wording of Baker, Nowakowski, Shalaby and Sharary [4], which [1] follows; the thesis [2] prints as a double Langford example the sequence $(4, 2, 3, 2, 4, 3, 4, 2, 3, 2, 4, 3)$, whose four 3s are the chain $3, 6, 9, 12$; and the published count 1 of two-fold Skolem sequences of order 1 [6] is the chain $(1, 1, 1, 1)$. Under the strict reading, which forbids chains, the in-bound cells $(d, l) = (1, 1)$, $(5, 4)$ and $(6, 6)$ would have no sequence, since each of their sequences contains a chain, and Theorem A would be false there. Definition 2.1 admits chains.

Coordinates. A cell (d, l) is *in-bound* if $3l \geq 2d - 1$, equivalently $2d \leq 3l + 1$. Its excess is $e = 3l - 2d + 1 \geq 0$. We use

$$\delta = d - l, \quad \mu = 2\delta - 1 = 2(d - l) - 1 = l - e.$$

In-bound means $\mu \leq l$, and $l \leq 2d - 1$ means $\mu \geq -l$. The line $\mu = l$ ($e = 0$) is the tight line; $\mu = -l$ ($e = 2l$) is the line $l = 2d - 1$.

Certificates. The formal development never builds a sequence directly. It builds coloured sets of integer pairs and proves once that they give a sequence.

Definition 2.4. An m -colour certificate for (m, d, l) is a family $C_0, \dots, C_{m-1}$ of finite sets of integer pairs (a, b) such that

- (i) for each colour c , the set $\{b - a : (a, b) \in C_c\}$ equals $[d, d + l - 1]$;
- (ii) the set of all endpoints of all pairs of all colours equals $[1, 2ml]$;
- (iii) $\sum_c |C_c| \leq ml$.

For $m = 2$ we write (A, B) for (C_0, C_1) and call it a two-colour certificate.

Lean: `L2.MultiPairing, L2.TwoFold, L2.TwoFold.toMulti (L2/Multi.lean).`

Lemma 2.5 (the bridge). *If an m -colour certificate exists for (m, d, l) with $d, l \geq 1$, then an m -fold Langford sequence of order l and defect d exists.*

Proof. Each colour has at least l pairs by (i). With (iii), each has exactly l , the colours are pairwise disjoint, and the difference map is injective on each colour, so each $p \in [d, d+l-1]$ is the difference of exactly m pairs. The ml pairs have at most $2ml$ endpoints, and by (ii) they have all of $[1, 2ml]$; so no endpoint is shared and the pairs partition $[1, 2ml]$ (a pair (a, a) is impossible because $d \geq 1$). Lemma 2.2 finishes. $\square$

Lean: L2.MultiPairing.card_colour, L2.MultiPairing.pairwise_disjoint,
L2.MultiPairing.card_fiber, L2.MultiPairing.endpoint_bij0n,
L2.MultiPairing.isLangford (L2/Bridge.lean).

The certificate format is the counting device of the formalization [14] lifted to m colours: families prove only set equalities and a cardinality bound, and the multiplicity bookkeeping is done in this one lemma. The pair machinery of [14] (endpoints, differences, translations) is reused, ported to the present toolchain.

3. EVERY MULTIPLICITY: THE BOUND, THE TIGHT LINE, THE ROW OF ORDER ONE

Theorem 3.1 (necessity). *Let $m, d, l \geq 1$. If an m -fold Langford sequence of order l and defect d exists, then*

$$2d + l \leq 2ml + 1, \quad \text{that is} \quad (2m - 1)l \geq 2d - 1,$$

with equality if and only if every pair has its left end in $[1, ml]$ and its right end in $[ml + 1, 2ml]$. If m is odd, moreover $l(2d + l + 1) \equiv 0 \pmod{4}$.

Proof. Take the pair form, put $H = ml$ and let L and R be the sets of left and right ends; they partition $[1, 2H]$ and have H elements each. Summing $b - a$ over the pairs,

$$\sum R - \sum L = m \sum_{p=d}^{d+l-1} p = \frac{ml(2d + l - 1)}{2}.$$

An H -subset of $[1, 2H]$ has sum at most $\sum[H+1, 2H]$ and at least $\sum[1, H]$, so $\sum R - \sum L \leq H^2 = m^2 l^2$, with equality only if $R = [H+1, 2H]$. Hence $2d + l - 1 \leq 2ml$, with the stated equality case. For the parity, $\sum R + \sum L = H(2H + 1)$, so

$$4 \sum R = 2ml(2ml+1) + ml(2d+l-1) = ml(4ml+2d+l+1) \equiv ml(2d+l+1) \pmod{4},$$

and the left side is divisible by 4. For odd m , m is invertible modulo 4. $\square$

Lean: Langford.necessary, L2.necessary_internal, L2.Nec.sum_le_top,
L2.Nec.bottom_le_sum, L2.Nec.sum_right_sub_left (L2/Necessity.lean); the equality
clause: Langford.tight_iff_straddle, L2.tight_iff_straddle_internal
(L2/Straddle.lean).

The equality case is formalized in the sequence form: for an m -fold Langford sequence s of order l and defect d , $2d + l = 2ml + 1$ if and only if $ml < i + s_i$ for every $i \in [1, ml]$. The two forms agree. If every pair straddles the midpoint, each $i \leq ml$ is a left end and its partner $i + s_i$ exceeds ml . Conversely, if $ml < i + s_i$ for every $i \leq ml$ and some $i \leq ml$ were a right end, its partner $a = i - s_i \geq 1$ would be a left end with $s_a = s_i$ and $a + s_a = i \leq ml$, contrary to the hypothesis at a ; so $[1, ml]$ consists of left ends, and since there are ml of them, the right ends are $[ml + 1, 2ml]$. The formal proof uses the same sums: at equality $2 \sum L = ml(ml + 1)$, which forces $L = [1, ml]$; conversely, if no right end lies in $[1, ml]$, then $R = [ml + 1, 2ml]$ and the distance sum gives $2d + l - 1 = 2ml$.

For even m the parity condition is vacuous (if l is odd, $2d + l + 1$ is even). For odd m it says: $l \equiv 0 \pmod{4}$ allows every d , $l \equiv 1 \pmod{4}$ needs d odd, $l \equiv 3 \pmod{4}$ needs d even, and $l \equiv 2 \pmod{4}$ is impossible. These are the classes of the ordinary case $m = 1$, where the two conditions are Simpson's [12]; at $d = 1$ they are the conditions of [4] for m -fold Skolem sequences, whose exactness there we take from the literature (the primary text of [4] was not available to us; its statement agrees in five secondary sources, among them Table 8 of [1]).

Remark 3.2. Nordh [10] studies *perfect Skolem sets*: a multiset of positive integers is perfect if $[1, 2n]$ can be partitioned into pairs whose differences are exactly its elements. A two-fold Langford sequence of order l and defect d is precisely a perfect multiset $2 \times [d, d + l - 1]$, and an m -fold one a perfect multiset $m \times [d, d + l - 1]$. Nordh's general necessary conditions — an even number of even elements, and for every k the k largest elements summing to at most $k(2n - k)$ — reduce for $m \times [d, d + l - 1]$, $n = ml$, exactly to Theorem 3.1. Indeed $f(k) = k(2n - k) - \sum_{i \leq k} a_i$ (elements in decreasing order) has increments decreasing by at least 1 at every step, so f is concave with $f(0) = 0$ and $f \geq 0$ everywhere if and only if $f(n) \geq 0$, which is $(2m - 1)l \geq 2d - 1$; and the number of even elements, m times the number of even $p \in [d, d + l - 1]$, is even for odd m exactly in the classes listed above. So his conditions add nothing to the counting bound; he notes himself that they are not sufficient for general multisets.

Status: proved here; not formalized (it is a comparison with the literature).

The tight line. Let $d_0 \geq 1$ and $l = 2d_0 - 1$. Table 1 of [1] is the ordinary Langford sequence of defect d_0 and order l with the pairs

$$(1) \quad (d_0 - r, 2d_0 + r) \ (0 \leq r \leq d_0 - 1), \quad (2d_0 - 1 - r, 3d_0 + r) \ (0 \leq r \leq d_0 - 2).$$

Their left ends fill $[1, l]$ and their right ends $[l + 1, 2l]$; the differences $d_0 + 2r$ and $d_0 + 1 + 2r$ fill $[d_0, 3d_0 - 2] = [d_0, d_0 + l - 1]$. As a bijection $i \mapsto (\text{right end}) - l$ of $[1, l]$ it is the two-block reversal τ_l of §4: it reverses $[1, d_0]$ and $[d_0 + 1, l]$.

Theorem 3.3 (the tight line, every m). *Let $m \geq 1$ and let $l = 2d_0 - 1$ be odd. Put $d = ((2m - 1)l + 1)/2 = ml - (d_0 - 1)$. The m colours*

$$C_c = \{ (a + cl, b + (m - 1 + c)l) : (a, b) \text{ a pair of } (1) \}, \quad 0 \leq c \leq m - 1,$$

form an m -colour certificate for (m, d, l) . Hence an m -fold Langford sequence exists on the whole line $(2m - 1)l = 2d - 1$.

Proof. Colour c has left ends $[cl + 1, (c + 1)l]$ and right ends $[ml + cl + 1, ml + (c + 1)l]$, so the union over c is $[1, 2ml]$. Its differences are those of (1) plus $(m - 1)l$, that is $[d_0 + (m - 1)l, d_0 + (m - 1)l + l - 1] = [d, d + l - 1]$. Each colour has l pairs. Lemma 2.5 finishes. When l is even, $(2m - 1)l + 1$ is odd and the line has no integer point. $\square$

Lean: `Langford.tight_exists`, `L2.table1`, `L2.tightColour`, `L2.tight_multi`,
`L2.tight_exists_internal` (`L2/Tight.lean`, `L2/Main.lean`).

By the equality case of Theorem 3.1, a tight sequence is the same thing as a permutation σ of $[1, ml]$, position i pairing with $ml + \sigma(i)$, whose displacement multiset $\{\sigma(i) - i\}$ is $m \times [-(l - 1)/2, (l - 1)/2]$. In the language of perfect sets this is Theorem 15 of [10] (numbering of arXiv:math/0506155v1), which should be cited for the bijection; the construction above is the block-diagonal permutation with m copies of τ_l . At $m = 2$ this closes the half $d \equiv 2 \pmod{6}$ of the tight line that Construction 2.7 of [1] does not reach.

Lemma 3.4 (juxtaposition). *If an m_1 -fold and an m_2 -fold Langford sequence of the same order l and defect d exist, so does an $(m_1 + m_2)$ -fold one: place the second after the first, shifted by $2m_1l$.*

Lean: L2.juxtapose, L2.MultiPairing.juxtapose (L2/Multi.lean).

Corollary 3.5. *For every even m , an m -fold Langford sequence of order l and defect d exists at every cell with $3l \geq 2d - 1$.*

Proof. Juxtapose $m/2$ copies of the two-fold sequence of Theorem A. □

Status: proved here from Theorem A and Lemma 3.4, both formalized; the corollary itself is not stated in the development. For odd $m \geq 3$ the same argument with one ordinary Langford sequence gives every cell with $l \geq 2d - 1$ that satisfies the parity condition, by Simpson's theorem [12], which is cited and not re-derived.

Theorem 3.6 (the row of order one). *Let $m, d \geq 1$. An m -fold Langford sequence of order 1 and defect d exists if and only if $d \mid m$.*

Proof. Here every pair has difference d and the positions are $[1, 2m]$. A position $i \leq d$ is a left end, since $i - d < 1$. A position $i > d$ is a right end exactly when $i - d$ is a left end, because the only possible partner of a right end i is $i - d$. By induction on i , the blocks $[1, d]$, $[d+1, 2d]$, $[2d+1, 3d]$, $\dots$ consist alternately of left ends and right ends. Write $2m = 2dq + r$ with $0 \leq r < 2d$. If $1 \leq r \leq d$, the position $2m$ lies in a block of left ends and its partner $2m + d$ is out of range. If $d < r < 2d$, the block $[2dq + 1, 2dq + d]$ consists of left ends and the partner $2dq + 2d$ of its last position exceeds $2m$. So $r = 0$ and $d \mid m$. Conversely, if $m = dk$, the cell $(d, 1)$ lies on the tight line of multiplicity d ($(2d - 1) \cdot 1 = 2d - 1$), so Theorem 3.3 gives a d -fold sequence, and k copies juxtaposed (Lemma 3.4) give an m -fold one. □

Lean: Langford.order_one_iff, L2.dvd_of_order_one, L2.order_one_of_dvd, L2.order_one_iff_internal (L2/OrderOne.lean).

A family of necessary conditions. For an integer T write

$$S(T) = \sum_{p=d}^{d+l-1} |p - T|$$

for the total distance from T to the differences.

Theorem 3.7 (the residue bound). *Let $m, d, l \geq 1$. If an m -fold Langford sequence of order l and defect d exists, then for every integer $T \geq 1$, with $r = ml \bmod T$,*

$$r(T - r) \leq m S(T).$$

Proof. Put $n = ml$ and take the pair form. For an integer y let $w(y) = T - \text{dist}(y, 4T\mathbb{Z})$, the triangle wave of period $4T$ with values in $[-T, T]$, and put $V(x) = w(2x - 2n - 1)$ for $x \in \mathbb{Z}$. Three properties of w are used: it is even; it is 1-Lipschitz, $|w(y) - w(y')| \leq |y - y'|$, as a distance to a set is; and $w(y + 2T) = -w(y)$, because shifting y by $2T$ turns its distance $\delta \in [0, 2T]$ to $4T\mathbb{Z}$ into $2T - \delta$. Hence $V(x + T) = -V(x)$ and $|V(x) - V(x')| \leq 2|x - x'|$.

The edge inequality. Let $(a, a + p)$ be a pair of the sequence. Then $V(a + p) = -V(a + p - T)$, so

$$|V(a) + V(a + p)| = |V(a) - V(a + p - T)| \leq 2|p - T|,$$

and for either sign $\sigma \in \{1, -1\}$, $\sigma (V(a) + V(a+p)) + 2|p-T| \geq 0$. Every position of $[1, 2n]$ is an end of exactly one pair, and every $p \in [d, d+l-1]$ is the difference of exactly m pairs, so the sum over the ml pairs is

$$(2) \quad \sigma \sum_{x=1}^{2n} V(x) + 2mS(T) \geq 0.$$

The total. As x runs over $[1, 2n]$, $2x - 2n - 1$ runs over $\pm 1, \pm 3, \dots, \pm(2n-1)$, and w is even, so $\sum_{x=1}^{2n} V(x) = 2g(n)$ with $g(N) = \sum_{k=0}^{N-1} w(2k+1)$. For $0 \leq k \leq T-1$ one has $0 < 2k+1 < 2T$, so $w(2k+1) = T-2k-1$, and $g(r) = \sum_{k < r} (T-2k-1) = r(T-r)$ for $0 \leq r \leq T$; in particular $g(T) = 0$, the terms $T-1, T-3, \dots, 1-T$ cancelling. Since $2(k+T)+1 = (2k+1) + 2T$, the terms of index $k+T$ are the negatives of those of index k , and $g(N+T) = g(T) - g(N) = -g(N)$. Writing $n = qT + r$ with $0 \leq r < T$,

$$\sum_{x=1}^{2n} V(x) = 2(-1)^q r(T-r).$$

The sign. With $\sigma = -(-1)^q$, (2) reads $-2r(T-r) + 2mS(T) \geq 0$. □

Lean: `Langford.residue_bound`, `L2.residue_bound_internal` (`L2/Residue.lean`).

In the formal statement $|p-T|$ is written $(p-T) + (T-p)$ with truncated subtraction, and $p = d+i$ with $0 \leq i < l$. The development follows the proof above: the triangle wave and its antiperiod, evenness and Lipschitz bound, the block sums g , the total over the positions, and the edge inequality summed over the pairs.

Remark 3.8. The potential of the proof is the certificate of Lemma 8.2 (§8): $V(x) = 2W_T(x-c)$ with $c = ml + \frac{1}{2}$, and $(y, z) = (\sigma V, 2|p-T|)$ is the member with $s = \sigma$. Theorem 3.7 says that the total of this certificate, with the sign chosen to minimize it, is nonnegative whenever a sequence exists; it holds for every $T \geq 1$, and in this form it is formalized. The summation uses only that every position is covered once and every p exactly m times, so, as in Lemma 8.1, the bound also holds for fractional pairings: a cell that violates it has an infeasible linear relaxation.

Status: proved here; the statement for fractional pairings is not formalized.

Corollary 3.9 (the window form). *If an m -fold Langford sequence of order l and defect d exists and $d \leq T \leq d+l-1$, then with $r = ml \bmod T$, $u = T-d$ and $v = d+l-1-T$,*

$$2r(T-r) \leq m(u(u+1) + v(v+1)).$$

Proof. For T in the window the distances $|p-T|$ are $u, u-1, \dots, 0$ for $p \leq T$ and $1, 2, \dots, v$ for $p > T$, so $2S(T) = u(u+1) + v(v+1)$. □

Lean: `L2.residue_bound_window` (`L2/Residue.lean`).

Corollary 3.10 (special cases). *In Theorem 3.7:*

- (i) for $T \geq \max(ml, d+l-1)$ the bound is the counting bound $2d+l \leq 2ml+1$ of Theorem 3.1;
- (ii) for $l=1$ and $T=d$ it says $d \mid m$, the necessity half of Theorem 3.6;
- (iii) for $T=d$ it says $r(d-r) \leq ml(l-1)/2$, where $r = ml \bmod d$.

Proof. (i) Every p is at most T , so $mS(T) = mlT - m \sum_p p$; and $r(T-r) = ml(T-ml)$, since $r = ml$ if $T > ml$ and both sides vanish if $T = ml$. The bound becomes $m \sum_p p \leq (ml)^2$, that is $ml(2d+l-1)/2 \leq (ml)^2$, that is $2d+l-1 \leq 2ml$. (ii) Here $S(d) = 0$ and $r = m \bmod d$, and $r(d-r) \leq 0$ with $0 \leq r < d$ forces $r = 0$. (iii) $S(d) = 0+1+\dots+(l-1) = l(l-1)/2$. $\square$

Status: proved here from Theorem 3.7; the corollary is not stated in the development, where the counting bound and the order-one theorem are proved directly (Theorems 3.1 and 3.6).

Theorem 3.11 (the forced-endpoint bound). *Let $m, d, l \geq 1$ and put $e = (2m-1)l - 2d + 1$. If an m -fold Langford sequence of order l and defect d exists, then*

$$mle \leq (l-1+e)^2, \quad \text{equivalently} \quad 6mld \leq 4d^2 + 2m^2l^2 + ml(l-1).$$

Proof. Put $n = ml$ and let L and R be the sets of left and right ends, as in the proof of Theorem 3.1. From $\sum R - \sum L = n(2d+l-1)/2$ and $\sum R + \sum L = n(2n+1)$,

$$4 \sum L = n(4n+3-2d-l).$$

By Theorem 3.1 and $l \geq 1, d \leq n$. Every position $x \in [1, d]$ is a left end, since a right end $a+p$ has $a \geq 1$ and $p \geq d$; and every left end a satisfies $a \leq 2n-d$, since $a+p \leq 2n$ and $p \geq d$. So L consists of $[1, d]$ and a set E of $n-d$ integers in $[d+1, 2n-d]$, and $\sum E$ is at most the sum of the $n-d$ largest integers of that interval, $[n+1, 2n-d]$:

$$2 \sum L \leq d(d+1) + (n-d)(3n-d+1).$$

With the value of $4 \sum L$ this becomes $0 \leq 4d^2 - 6nd + 2n^2 + n(l-1)$, the second form. Since $l-1+e = 2(n-d)$, the right side equals $(l-1+e)^2 - ne$. $\square$

Lean: `Langford.forced.endpoint, L2.forced.endpoint.internal (L2/Forced.lean)`.

The development states the second form as $6mld + ml \leq 4d^2 + 2(ml)^2 + ml^2$. The bound is the case $T = d$ of Theorem 3.7, weakened. Write $n = ml = qd + r$ with $0 \leq r < d$; then

$$(n-d)(n-2d) + r(d-r) = (1-q)d((2-q)d-2r) \geq 0,$$

since for $q = 0$ both factors are positive, for $q = 1$ the first vanishes, and for $q \geq 2$ both are at most 0. With Corollary 3.10(iii), $ml(l-1)/2 \geq r(d-r) \geq -(n-d)(n-2d)$, and $2(n-d)(n-2d) + ml(l-1) = 4d^2 - 6nd + 2n^2 + n(l-1)$. In the notation of §8 the forced-endpoint bound is $2F_1(d) \geq 0$, and the inequality used is the case $j = 1, T = d$ of (3). The direct proof above is the one formalized; it needs only the endpoint sets and their sums.

4. THE REDUCTION AT $m = 2$

Concatenation and the unsplittable region.

Lemma 4.1 (concatenation). *If two-fold sequences of orders l_1, l_2 and defects d and $d+l_1$ exist, then one of order l_1+l_2 and defect d exists: the second, shifted by $4l_1$, follows the first.*

Proof. The differences are $[d, d+l_1-1] \cup [d+l_1, d+l_1+l_2-1]$, twice each, and the positions are $[1, 4l_1] \cup [4l_1+1, 4l_1+4l_2]$. On certificates the two colours are concatenated colour by colour. $\square$

Lean: `L2.TwoFold.concat (L2/Multi.lean)`.

Lemma 4.2 (splitting). *Let (d, l) be in-bound. A split $l = l_1 + l_2$ with both (d, l_1) and $(d + l_1, l_2)$ in-bound exists if and only if $e \geq 5l_1^*$, where $l_1^* = \lceil (2d - 1)/3 \rceil = \lfloor (2d + 1)/3 \rfloor$, and then $l_1 = l_1^*$ works. Every in-bound cell with $l \geq 2d$ splits with $l_1 = l_1^*$, and every cell that does not split has $l < (16d + 2)/9 \leq 2d$.*

Proof. (d, l_1) is in-bound iff $3l_1 \geq 2d - 1$, and $(d + l_1, l - l_1)$ is in-bound iff $2(d + l_1) \leq 3(l - l_1) + 1$, that is $5l_1 \leq e$. The first is a lower bound on l_1 and the second an upper bound, so a split exists iff the least admissible l_1^* satisfies the second. If $l \geq 2d$ then $e \geq 4d + 1 \geq (10d + 5)/3 \geq 5l_1^*$ and $1 \leq l_1^* < l$. If $e < 5l_1^* \leq 5(2d + 1)/3$ then $9l < 16d + 2$. $\square$

Lean: used inside `L2.twoFold.all` (`L2/Main.lean`).

The unsplittable region $0 \leq e < 5\lceil (2d - 1)/3 \rceil$ is therefore a wedge of slope roughly $[2/3, 16/9]$ in l/d , and the counting bound shows that no concatenation of consecutive blocks can reach it. The cells with $l \geq 2d - 1$ all split except $(d, l) = (1, 1)$ and $(4, 7)$: at $l = 2d - 1$ one needs $5l_1^* \leq 4d - 2$, which holds for $d \geq 6$ since $5(2d + 1)/3 \leq 4d - 2$, and for $d \leq 5$ fails only at $d = 1$ and $d = 4$. Both exceptional cells lie on the line $\mu = -l$ and are reached through Lemma S by τ_l^{-1} , so Simpson's theorem is not needed anywhere.

The signed-permutation problem.

Definition 4.3. Let $l \geq 1$ and $\delta \in \mathbb{Z}$. A permutation σ of $[1, l]$ solves $\text{SP}(l, \delta)$ if the $2l$ numbers

$$y_w = \sigma(w) - w + \delta \quad \text{and} \quad 1 - y_w \quad (w \in [1, l])$$

are exactly the elements of $[1 - l, l]$. In the formalization σ is its graph $G \subset \mathbb{Z}^2$: the first coordinates of G are $[1, l]$, the second coordinates are $[1, l]$, $|G| \leq l$, and the images of G under the *value* $(w, t) \mapsto t - w + \delta$ and the *mirror* $(w, t) \mapsto w - t + 1 - \delta$ have union $[1 - l, l]$.

Lean: `L2.SP`, `L2.val`, `L2.mir` (`L2/SP.lean`, `L2/Rows.lean`).

Lemma 4.4 (displacement form). *Put $\mu = 2\delta - 1$ and $x_w = 2(\sigma(w) - w) + \mu$. Then σ solves $\text{SP}(l, \delta)$ if and only if $\{|x_w| : w \in [1, l]\} = \{1, 3, \dots, 2l - 1\}$.*

Proof. $x_w = 2y_w - 1$, and for an integer y one has $y \in \{k, 1 - k\}$ exactly when $|2y - 1| = 2k - 1$. The set $[1 - l, l]$ is the disjoint union of the l pairs $\{k, 1 - k\}$, $k \in [1, l]$, and $y_w, 1 - y_w$ always lie in the same pair. So the $2l$ numbers are $[1 - l, l]$ iff every pair is hit by exactly one w , iff the $|x_w|$ run through $2k - 1$, $k \in [1, l]$, once each. $\square$

Lean: implicit in `L2.SP` (the set form is the one formalized).

We call $k = (|x_w| + 1)/2$ the *class* of w . In the form in which the construction was first found, a solution was a permutation π of $[1, l]$ with a label S or D on each u , the value of u being $u + \pi(u) - 1$ (label S) or $\pi(u) - u$ (label D), the l values forming $[\delta, \delta + l - 1]$; the two forms correspond by $u = (|x_w| + 1)/2$, $\pi(u) = w$, the label being the sign of x_w .

Lemma 4.5 (Lemma S). *Let σ solve $\text{SP}(l, \delta)$ and suppose $d = l + \delta \geq 1$. With $y_w = \sigma(w) - w + \delta$, the pairs*

$$A_w = (l + 1 - w, 2l + y_w), \quad B_w = (2l + 1 - y_w, 3l + w), \quad w \in [1, l],$$

form a two-colour certificate $(\{A_w\}, \{B_w\})$ for $(2, d, l)$. The resulting two-fold Langford sequence of order l and defect d is symmetric under $i \mapsto 4l + 1 - i$, which exchanges A_w and B_w , and every pair joins the outer half $[1, l] \cup [3l + 1, 4l]$ to the middle half $[l + 1, 3l]$.

Proof. The difference of A_w is $2l + y_w - l - 1 + w = d + \sigma(w) - 1$, and so is the difference of B_w . As $\sigma(w)$ runs over $[1, l]$, each colour has the difference set $[d, d + l - 1]$. The left ends of the A_w are $[1, l]$ and the right ends of the B_w are $[3l + 1, 4l]$. The remaining endpoints are $2l + y_w$ and $2l + (1 - y_w)$, which by the definition of SP run over $2l + [1 - l, l] = [l + 1, 3l]$. So the endpoints are $[1, 4l]$, and there are $2l$ pairs. The symmetry is the identity $4l + 1 - (2l + y_w) = 2l + 1 - y_w$ together with $4l + 1 - (l + 1 - w) = 3l + w$. $\square$

Lean: L2.colourA, L2.colourB, L2.SP.toTwoFold (L2/LemmaS.lean).

Not every cell has a sequence of this form. The next invariant shows that exactly one small cell of each sign is missed.

Lemma 4.6 (the value-sum invariant). *Let σ solve $\text{SP}(l, \delta)$, let K^+ be the set of classes k with $y_w = k$ for some w and K^- the set with $y_w = 1 - k$. Then*

$$\sum_{k \in K^-} (2k - 1) = \frac{l(l - \mu)}{2} = \frac{le}{2}.$$

In particular $\text{SP}(4, 2)$ ($\mu = 3$) and $\text{SP}(4, -1)$ ($\mu = -3$) have no solution.

Proof. $\sum_w y_w = \sum \sigma(w) - \sum w + l\delta = l\delta$. On the other hand $K^+ \sqcup K^- = [1, l]$ and $\sum_w y_w = \sum_{K^+} k + \sum_{K^-} (1 - k) = l(l + 1)/2 - \sum_{K^-} (2k - 1)$. So $\sum_{K^-} (2k - 1) = l(l + 1)/2 - l\delta = l(l - \mu)/2$. At $(l, \mu) = (4, 3)$ this is 2; at $(4, -3)$ it is 14, so $\sum_{K^+} (2k - 1) = 16 - 14 = 2$. No set of distinct odd numbers sums to 2. $\square$

Status: proved here; not formalized. The formal development never needs these two cells: it uses literal sequences at $(d, l) = (6, 4)$ and $(3, 4)$ and the base $(7, 3)$ in Theorem 4.11.

The two cells are $(d, l) = (6, 4)$ and $(3, 4)$. The first has no mirror-symmetric two-fold sequence at all (the same invariant, applied after folding a symmetric sequence at its centre, where no pair can be its own mirror image); it does have the sequence

$$(6, 9, 9, 6, 8, 8, 6, 7, 7, 6, 9, 9, 8, 8, 7, 7).$$

The second has a mirror-symmetric sequence that is not of the form of Lemma S. Both are among the sixteen literal cells of §6.

Symmetry, the reversal, τ_l and Lemma C.

Lemma 4.7 (inversion). *σ solves $\text{SP}(l, \delta)$ if and only if σ^{-1} solves $\text{SP}(l, 1 - \delta)$; in displacement terms $\mu \mapsto -\mu$.*

Proof. The graph of σ^{-1} consists of the pairs (t, w) . The value of (t, w) at $1 - \delta$ is $w - t + 1 - \delta$, the mirror of (w, t) at δ , and its mirror at $1 - \delta$ is $t - w + \delta$, the value of (w, t) at δ . So the union of values and mirrors is unchanged. $\square$

Lean: L2.SP.inv, L2.mem_image_val_swap, L2.mem_image_mir_swap.

A second proof uses the reflection $w \mapsto l + 1 - w$ on the targets with the labels S and D exchanged; it is a different bijection between the two solution sets.

Lemma 4.8 (the reversal and τ_l). *The reversal $w \mapsto l + 1 - w$ solves $\text{SP}(l, 1)$ ($\mu = 1$) for every $l \geq 1$. For odd l , the permutation τ_l that reverses $[1, (l + 1)/2]$ and $[(l + 3)/2, l]$ has displacements $\tau_l(i) - i$ running through $[-(l - 1)/2, (l - 1)/2]$ once each and solves $\text{SP}(l, (l + 1)/2)$ ($\mu = l$); hence τ_l^{-1} solves $\mu = -l$.*

Proof. For the reversal, $y_w = l + 2 - 2w$ runs through $l, l - 2, \dots, 2 - l$ and $1 - y_w$ through the integers of the other parity in $[1 - l, l - 1]$. For τ_l with $h = (l + 1)/2$: on $[1, h]$ the displacement is $h + 1 - 2i$, running over $h - 1, h - 3, \dots, 1 - h$; on $[h + 1, l]$ it is $l + h + 1 - 2i$, running over $h - 2, h - 4, \dots, 2 - h$. Together they are $[1 - h, h - 1]$ once each, so $y = (\text{displacement}) + h$ runs through $[1, l]$ and the mirrors through $[1 - l, 0]$. $\square$

Lean: L2.SP.reversal, L2.tauGraph, L2.SP.tau.

Lemma 4.9 (Lemma C, with inversion). *Let σ_0 solve $\text{SP}(n, \delta)$ with $\delta \geq 1$, and put $\mu = 2\delta - 1$. Define a permutation σ_1 of $[1, n + \mu]$ by*

$$\sigma_1(w) = \sigma_0(w) + \mu \quad (1 \leq w \leq n), \quad \sigma_1(n + i) = \tau_\mu(i) \quad (1 \leq i \leq \mu).$$

Then σ_1 solves $\text{SP}(n + \mu, 1 - \delta)$, and σ_1^{-1} solves $\text{SP}(n + \mu, \delta)$.

Proof. The images are $[\mu + 1, n + \mu]$ and $[1, \mu]$, so σ_1 is a permutation. For $w \leq n$, the value of $(w, \sigma_0(w) + \mu)$ at $1 - \delta$ is $\sigma_0(w) - w + \mu + 1 - \delta = \sigma_0(w) - w + \delta$, the old value, and its mirror is likewise the old mirror; these contribute $[1 - n, n]$. For the appended block, the value of $(n + i, \tau_\mu(i))$ at $1 - \delta$ is $(\tau_\mu(i) - i) - n + 1 - \delta$; the displacements of τ_μ are $[1 - \delta, \delta - 1]$ (Lemma 4.8), so these values are $[1 - (n + \mu), -n]$ and their mirrors $[n + 1, n + \mu]$. The union is $[1 - (n + \mu), n + \mu]$. Lemma 4.7 gives the last claim. No bound on μ in terms of n is used. $\square$

Lean: L2.cinvGraph, L2.SP.cinv_aux, L2.SP.cinv.

Corollary 4.10 (descent). *If $\text{SP}(n, \delta)$ has a solution and $\delta \geq 1$, then so has $\text{SP}(n + k\mu, \delta)$ for every $k \geq 0$, where $\mu = 2\delta - 1$.*

Lean: L2.SP.descend.

In its first form Lemma C carried (l, μ) to $(l + |\mu|, -\mu)$; composed with inversion it holds μ fixed and walks l up the whole line of fixed μ . An equivalent composite, written with the reflection, gives the same extension.

Reduction to the band.

Theorem 4.11 (Theorem R). *Assume the band theorem (Theorem 5.7): for every $l \geq 5$ and every odd μ with $l < 2\mu \leq 2l$, $\text{SP}(l, (\mu + 1)/2)$ has a solution. Then $\text{SP}(l, \delta)$ has a solution for every $l \geq 5$ and every δ with $-l \leq 2\delta - 1 \leq l$.*

Proof. By Lemma 4.7 we may assume $\mu = 2\delta - 1 \geq 1$. We use strong induction on l . If $\mu = 1$, use the reversal. If $l < 2\mu$, use the band theorem (it includes $\mu = l$, which is τ_l). Otherwise $2\mu \leq l$; put $l' = l - \mu \geq \mu$ and obtain a solution at (l', μ) as follows, then apply Lemma 4.9.

- If $l' = \mu$: τ_μ .
- If $\mu < l' < 2\mu$ and $l' \geq 5$: the band theorem.
- If $\mu < l' < 2\mu$ and $l' \leq 4$: then μ is odd, $\mu \geq 3$ and $\mu < l' \leq 4$, so $(l', \mu) = (4, 3)$ and $l = 7$. That cell is empty (Lemma 4.6), and $(7, 3)$ is supplied directly by $\sigma = (6, 4, 7, 3, 5, 2, 1)$, whose values $|x_w|$ are 13, 7, 11, 1, 3, 5, 9.
- If $l' \geq 2\mu$: then $l' \geq 6$, and the induction hypothesis applies. $\square$

Lean: L2.cone_sp, L2.sp_7.3 (L2/Cone.lean, L2/Literals.lean).

Equivalently, every cell descends along its line of fixed μ to the base $l_0 = \mu + ((l - \mu) \bmod \mu)$, which lies in $[\mu, 2\mu)$, except that the base $(4, 3)$ is replaced by $(7, 3)$. The inversion and descent steps are proved in general; everything specific sits in the band.

5. THE BAND

The band is the set of cells (l, μ) with $l \geq 5$, μ odd and $l/2 < \mu \leq l$. All solutions used there are *block reversals*.

Shapes and the block lemma. A *shape* is a list of block sizes $n_0, \dots, n_{b-1} \geq 1$ with sum l and a *target order*, a permutation of the block indices listing them in left-to-right target position. Block i has source interval $[a_i + 1, a_i + n_i]$ with $a_i = n_0 + \dots + n_{i-1}$ (sources cut left to right) and target interval $[c_i + 1, c_i + n_i]$, where c_i is the sum of the sizes of the blocks placed before i in the target order; the block maps its source interval onto its target interval reversed:

$$\sigma(a_i + r) = c_i + n_i + 1 - r, \quad 1 \leq r \leq n_i.$$

In the formalization block i is the finite set $\text{row}(a_i, c_i, n_i) = \{(a_i + r, c_i + n_i + 1 - r) : 1 \leq r \leq n_i\}$ and a family is a union of rows.

Lemma 5.1 (block lemma). *Put $M_i = 2(c_i - a_i) + \mu$. On block i , $x_{a_i+r} = M_i + 2n_i + 2 - 4r$ for $1 \leq r \leq n_i$: an arithmetic progression of step 4 from $x_{l_0} = M_i - 2n_i + 2$ to $x_{h_i} = M_i + 2n_i - 2$. Consequently its classes form a run of one parity and step 2:*

- *type P* ($x_{l_0} \geq 1$): classes $(M_i - 2n_i + 3)/2, \dots, (M_i + 2n_i - 1)/2$;
- *type N* ($x_{h_i} \leq -1$): classes $(-M_i - 2n_i + 3)/2, \dots, (-M_i + 2n_i - 1)/2$;
- *type C* ($x_{l_0} \leq -1, x_{h_i} \geq 1$): if $x_{h_i} \equiv 1 \pmod{4}$ (type C1), odd classes $1, \dots, (x_{h_i} + 1)/2$ and even classes $2, \dots, (1 - x_{l_0})/2$; if $x_{h_i} \equiv 3 \pmod{4}$ (type C3), even classes $2, \dots, (x_{h_i} + 1)/2$ and odd classes $1, \dots, (1 - x_{l_0})/2$.

Proof. $x_{a_i+r} = 2(c_i + n_i + 1 - r - a_i - r) + \mu$. The classes follow from $k = (|x| + 1)/2$. In the set form of Definition 4.3, the values of the block run with step 2 from $c_i - a_i + 1 - n_i + \delta$ to $c_i - a_i + n_i - 1 + \delta$, and the mirrors from $a_i - c_i + 2 - n_i - \delta$ to $a_i - c_i + n_i - \delta$. $\square$

Lean: L2.row, L2.mem_image_val_row, L2.mem_image_mir_row, L2.mem_image_fst_row, L2.mem_image_snd_row (L2/Rows.lean).

Lemma 5.2 (tiling). *Suppose that the odd class runs of a shape, in increasing order, start at 1, each starts 2 above the end of the previous one, and the last ends at the largest odd number $\leq l$; and that the even runs do the same from 2 to the largest even number $\leq l$. Then the shape solves $\text{SP}(l, (\mu + 1)/2)$.*

Proof. The classes $1, \dots, l$ then occur once each, and Lemma 4.4 applies; the shape is a permutation because the blocks tile $[1, l]$ both in source order and in target order. $\square$

A *family* is a target order together with sizes n_i that are affine functions of (l, μ) on a residue class, and a *domain*. Its *class table* lists, for each block, n_i , M_i , the type and the class run(s), and the order of the runs in the odd and in the even chain. A family is proved by four facts, each an identity or a linear inequality in (l, μ) : (a) the sizes sum to l ; (b) the chain relations (first start, abutment, last end) hold identically; (c) the domain is exactly the set where every $n_i \geq 1$ and every type condition holds; (d) on the residue class every

size and run end is an integer and each C block has a constant residue of x_{hi} modulo 4. Then at every cell of the domain Lemma 5.2 applies.

The class tables of all families are printed in Appendix A. Each was derived symbolically from its sizes and target order, and each was re-derived by an independent implementation: for the even classes of l on every residue sublattice modulo 32, for the odd classes with exact linear programming proving the sign conditions over the domain and the exactness of the domain in both directions. The two checkers of §9 transcribe every table independently. Every family was, in addition, built at every domain cell up to at least $l = 400$, and at domain-edge cells up to $l = 1001$ or beyond, and checked through Lemma S to a two-fold sequence.

The families. Table 1 lists the families. There are twenty-one block-reversal families and τ_l (the two-block family TT), twenty-two in all.

TABLE 1. The band families. Domains in full, with class tables, in Appendix A.

family	l	μ	blocks	range of μ
FA	$\equiv 2 \pmod{4}$	odd	5	$(l+4)/2 \leq \mu \leq (3l-2)/4$
FB2	$\equiv 2 \pmod{8}$	odd	5	$(3l+6)/4 \leq \mu \leq l-1$
FB6	$\equiv 6 \pmod{8}$	odd	5	$(3l+10)/4 \leq \mu \leq l-1$
F4	$\equiv 6 \pmod{8}$	$(3l+2)/4$	4	a line
A1	$\equiv 0 \pmod{4}$	$\equiv 1 \pmod{4}$	9	$l/2 + 1 \leq \mu \leq 3l/4 - 2$
A3	$\equiv 0 \pmod{4}$	$\equiv 3 \pmod{4}$	9	$l/2 + 1 \leq \mu \leq 3l/4 - 2, \mu \leq l-5$
U1	$\equiv 0 \pmod{4}$	$\equiv 1 \pmod{8}$	8	$(2l+11)/3 \leq \mu \leq l-3$
U3	$\equiv 0 \pmod{4}$	$\equiv 3 \pmod{8}$	9	$(2l+1)/3 \leq \mu \leq l-3$
U5	$\equiv 0 \pmod{4}$	$\equiv 5 \pmod{8}$	8	$(2l+7)/3 \leq \mu \leq l-3$
U7	$\equiv 0 \pmod{4}$	$\equiv 7 \pmod{8}$	9	$(4l+11)/5 \leq \mu \leq l-1$
V	$\equiv 0 \pmod{4}$	$\equiv 3 \pmod{4}$	9	about $(2l+9)/3 \leq \mu \leq (4l+3)/5$
T	$\equiv 0 \pmod{4}$	$\equiv 3 \pmod{8}$	7	$(8l+11)/9 \leq \mu \leq l-1$
FA3	$\equiv 1 \pmod{4}$	$\equiv 3 \pmod{4}$	7	$(l+5)/2 \leq \mu \leq l-6$
FA1	$\equiv 1 \pmod{4}$	$\equiv 1 \pmod{4}$	8	$(l+5)/2 \leq \mu \leq l-4$
FT	$\equiv 1 \pmod{4}$	$l-2$	6	a line, $l \geq 9$
FL1	$\equiv 1 \pmod{8}$	$(l+1)/2$	7	a line, $l \geq 17$
FL5	$\equiv 5 \pmod{8}$	$(l+1)/2$	6	a line, $l \geq 13$
R3O	$\equiv 3 \pmod{4}$	$\equiv 1 \pmod{4}$	7	$(l+3)/2 \leq \mu \leq l-6$
R3E	$\equiv 3 \pmod{4}$	$\equiv 3 \pmod{4}$	8	$(l+7)/2 \leq \mu \leq l-4$
R3Q	$\equiv 3 \pmod{4}$	$\equiv 3 \pmod{4}$	7	$(l+3)/2 \leq \mu \leq (2l-1)/3$
R3T1	$\equiv 3 \pmod{4}$	$l-2$	6	a line, $l \geq 7$
TT	odd	l	2	τ_l

The families for $l \equiv 0 \pmod{4}$ were found by enumerating the block decompositions of computed solutions and solving, for each solution, the linear system that its pattern of abutting runs imposes on the sizes; a single solution then yields its whole two-parameter family. The other classes were found the same way from shape enumerations. Two structural features recur. In R3O, R3E, R3Q and all the $l \equiv 1 \pmod{4}$ families a *corner* block maps the last $(l-\mu)/2$ sources reversed onto the first $(l-\mu)/2$ targets, with $M = -l$, and is the only block of type N. In R3O, R3E and R3Q a *double block*, two blocks consecutive in source and in target that share one M , covers the bottom run of both parities.

Sporadic cells and bases. Table 2 lists the eleven explicit solutions. The eight cells with $l \equiv 0 \pmod{4}$ are exactly the band cells of that class that lie in no family domain; $(5, 3)$ and $(9, 5)$ complete the class $l \equiv 1 \pmod{4}$; $(7, 3)$ is the base of Theorem 4.11.

TABLE 2. The explicit solutions, $\sigma = (\sigma(1), \dots, \sigma(l))$, $\delta = (\mu + 1)/2$.

(l, μ)	δ	σ
$(7, 3)$	2	6, 4, 7, 3, 5, 2, 1
$(5, 3)$	2	3, 5, 2, 4, 1
$(9, 5)$	3	6, 5, 9, 8, 1, 7, 4, 3, 2
$(8, 5)$	3	2, 6, 8, 7, 5, 4, 3, 1
$(8, 7)$	4	1, 5, 4, 8, 7, 2, 6, 3
$(12, 9)$	5	7, 6, 5, 1, 12, 11, 10, 4, 9, 8, 3, 2
$(12, 11)$	6	2, 1, 3, 9, 8, 12, 11, 10, 7, 6, 5, 4
$(16, 11)$	6	1, 4, 12, 11, 10, 16, 15, 14, 13, 9, 8, 7, 6, 5, 3, 2
$(20, 15)$	8	2, 1, 14, 13, 12, 11, 10, 20, 19, 18, 17, 16, 15, 7, 6, 5, 4, 3, 9, 8
$(24, 17)$	9	2, 1, 17, 16, 15, 14, 13, 12, 24, 23, 22, 21, 20, 19, 18, 9, 8, 7, 6, 5, 4, 3, 11, 10
$(32, 23)$	12	2, 1, 23, 22, 21, 20, 19, 18, 17, 16, 15, 3, 32, 31, 30, 29, 28, 27, 26, 25, 24, 11, 10, 9, 8, 7, 6, 5, 4, 14, 13, 12

Lean: L2.sp_7_3, L2.sp_5_3, L2.sp_9_5, L2.sp_8_5, L2.sp_8_7, L2.sp_12_9, L2.sp_12_11, L2.sp_16_11, L2.sp_20_15, L2.sp_24_17, L2.sp_32_23 (L2/Literals.lean, each by kernel evaluation of a literal finite-set equality).

The coverage theorems.

Theorem 5.3 ($l \equiv 2 \pmod{4}$). *Every band cell with $l \equiv 2 \pmod{4}$, $l \geq 6$, has a solution.*

Proof. Here $\mu = l$ is even and not in the band, so μ runs over the odd numbers in $(l/2, l)$. If $l = 8k + 2$ ($k \geq 1$), these are $[4k + 3, 8k + 1]$: FA covers $[4k + 3, 6k + 1]$ and FB2 covers $[6k + 3, 8k + 1]$. If $l = 8k + 6$, they are $[4k + 5, 8k + 5]$: FA covers $[4k + 5, 6k + 3]$, F4 is $\mu = 6k + 5$, and FB6 covers $[6k + 7, 8k + 5]$. In each case the stated interval is the family's domain restricted to the class. $\square$

Lean: L2.band_r2 (L2/Band.lean), with L2.famFA_sp, L2.famFB2_sp, L2.famFB6_sp, L2.famF4_sp (L2/FamR2.lean).

Theorem 5.4 ($l \equiv 1 \pmod{4}$). *Every band cell with $l \equiv 1 \pmod{4}$, $l \geq 5$, has a solution.*

Proof. The odd μ in $(l/2, l]$ are $(l + 1)/2$, which is odd, and then $(l + 5)/2, \dots, l$, since $(l + 3)/2$ is even. The case $\mu = (l + 1)/2$ is FL1 if $l \equiv 1 \pmod{8}$, $l \geq 17$; FL5 if $l \equiv 5 \pmod{8}$, $l \geq 13$; and the bases $(5, 3)$ and $(9, 5)$ at $l = 5$ and 9 . For $(l + 5)/2 \leq \mu \leq l - 4$: FA1 if $\mu \equiv 1 \pmod{4}$; if $\mu \equiv 3 \pmod{4}$, then $\mu \leq l - 6$ because $l - 4 \equiv 1 \pmod{4}$, and FA3 applies. The case $\mu = l - 2$ is FT for $l \geq 9$ and the base $(5, 3)$ at $l = 5$. The case $\mu = l$ is τ_l . $\square$

Lean: L2.band_r1, with L2.famFA1_sp, L2.famFA3_sp, L2.famFT_sp, L2.famFL1_sp, L2.famFL5_sp (L2/FamR1.lean).

Theorem 5.5 ($l \equiv 3 \pmod{4}$). *Every band cell with $l \equiv 3 \pmod{4}$, $l \geq 7$, has a solution.*

Proof. Write $l = 4q + 3$ and $t = (l - \mu)/2$; the band is $0 \leq t \leq q$. The case $t = 0$ is τ_l ; $t = 1$ is R3T1; odd t with $3 \leq t \leq q$ is R3O; even t with $2 \leq t \leq q - 1$ is R3E. What remains is $t = q$ with q even, so $q \geq 2$; then $3t = 3q \geq 2q + 2$, and R3Q applies (its domain in these coordinates is t even, $2q + 2 \leq 3t, t \leq q$). $\square$

Lean: L2.band_r3, with L2.famR3O_sp, L2.famR3E_sp, L2.famR3Q_sp, L2.famR3T1_sp (L2/FamR3.lean).

Theorem 5.6 ($l \equiv 0 \pmod{4}$). *Every band cell with $l \equiv 0 \pmod{4}$, $l \geq 8$, has a solution.*

Proof. The band consists of the odd μ with $l/2 + 1 \leq \mu \leq l - 1$ ($l/2$ is even). For $l \geq 68$ use the *main intervals*

$$\begin{array}{ll} \text{A1, A3:} & [l/2 + 1, 3l/4 - 2], \quad \text{U1:} \quad [(2l + 11)/3, l - 3], \\ \text{U3:} & [(2l + 1)/3, l - 3], \quad \text{U5:} \quad [(2l + 7)/3, l - 3], \\ \text{U7:} & [(4l + 11)/5, l - 1], \quad \text{V:} \quad [(2l + 9)/3, (4l + 3)/5], \\ \text{T:} & [(8l + 11)/9, l - 1]. \end{array}$$

Each domain inequality is linear and holds at both ends of the main interval for $l \geq 68$, so every μ of the family's class in its main interval lies in its domain. Split by $\mu \bmod 8$.

- $\mu \equiv 1$ or 5 : A1, then U1 or U5, up to $l - 3$. The intervals overlap because $(2l + 11)/3 \leq 3l/4 - 2$ iff $l \geq 68$ and $(2l + 7)/3 \leq 3l/4 - 2$ iff $l \geq 52$; and $l - 1$ is never $\equiv 1$ or $5 \pmod{8}$.
- $\mu \equiv 3$: A3, then U3 ($(2l + 1)/3 \leq 3l/4 - 2$ iff $l \geq 28$) up to $l - 3$; the cell $\mu = l - 1$ (when $l \equiv 4 \pmod{8}$) is in T, since $(8l + 11)/9 \leq l - 1$ iff $l \geq 20$.
- $\mu \equiv 7$: A3, then V ($(2l + 9)/3 \leq 3l/4 - 2$ iff $l \geq 60$), then U7 up to $l - 1$. A cell strictly between $(4l + 3)/5$ and $(4l + 11)/5$ would have $5\mu - 4l \in [4, 10]$; but $5\mu - 4l \equiv 3 \pmod{8}$, since $\mu \equiv 7 \pmod{8}$ and $4l \equiv 0 \pmod{16}$, so there is none.

For $8 \leq l \leq 64$ every band cell was checked against the family domains: all lie in some domain except $(l, \mu) = (8, 5), (8, 7), (12, 9), (12, 11), (16, 11), (20, 15), (24, 17)$ and $(32, 23)$, which are the sporadic cells of Table 2. This finite part is a decision of linear arithmetic for each of the fifteen values $l/4 \in [2, 16]$, and the formal proof makes it exactly so. $\square$

Lean: L2.band_r0, with L2.famA1_sp, L2.famA3_sp, L2.famU1_sp, L2.famU3_sp (L2/FamR0a.lean) and L2.famU5_sp, L2.famU7_sp, L2.famV_sp, L2.famT_sp (L2/FamR0b.lean).

The coverage in this class was also re-proved for every l , independently, by exact lattice geometry (elimination with residue tightening and enumeration of the bounded remainder), with the same eight sporadic cells as the only uncovered ones.

Theorem 5.7 (the band). *For every $l \geq 5$ and every odd μ with $l/2 < \mu \leq l$, $\text{SP}(l, (\mu+1)/2)$ has a solution.*

Proof. Split by $l \bmod 4$ and apply Theorems 5.3, 5.4, 5.5 and 5.6; for even l the band stops at $\mu = l - 1$. $\square$

Lean: L2.band_sp (L2/Band.lean).

Together with Theorem 4.11, Theorem 5.7 gives $\text{SP}(l, \delta)$ on the whole cone $-l \leq 2\delta - 1 \leq l$ for every $l \geq 5$.

6. PROOF OF THEOREM A

Proof of Theorem A. Necessity is Theorem 3.1 at $m = 2$: $2d + l \leq 4l + 1$. For sufficiency we prove by strong induction on l that every in-bound cell (d, l) has a two-colour certificate, and conclude by Lemma 2.5.

- (i) If $l \leq 4$, the cell is one of the sixteen in-bound cells of Table 3, each an explicit sequence.
- (ii) If $l \geq 5$ and $l \geq 2d$, Lemma 4.2 splits it with $l_1 = \lfloor (2d + 1)/3 \rfloor$ into two in-bound cells of smaller order; the induction hypothesis gives both, and Lemma 4.1 joins them.
- (iii) If $l \geq 5$ and $l \leq 2d - 1$, put $\delta = d - l$ and $\mu = 2\delta - 1$. Then $\mu \leq l$ because the cell is in-bound, and $\mu \geq -l$ because $l \leq 2d - 1$. Theorem 4.11 with Theorem 5.7 gives a solution of $\text{SP}(l, \delta)$, and Lemma 4.5 gives a certificate for $(d, l) = (l + \delta, l)$. $\square$

Lean: `Langford.twoFold.exists_iff, L2.twoFold.all, L2.twoFold.exists_iff_internal (L2/Main.lean).`

TABLE 3. The sixteen in-bound cells with $l \leq 4$: one two-fold sequence each, as its $2l$ pairs of positions.

(d, l)	pairs
(1, 1)	[1, 2], [3, 4]
(2, 1)	[1, 3], [2, 4]
(1, 2)	[1, 2], [3, 5], [4, 6], [7, 8]
(2, 2)	[1, 3], [2, 5], [4, 7], [6, 8]
(3, 2)	[1, 4], [2, 6], [3, 7], [5, 8]
(1, 3)	[1, 2], [3, 4], [5, 7], [6, 9], [8, 11], [10, 12]
(2, 3)	[1, 4], [2, 6], [3, 7], [5, 8], [9, 11], [10, 12]
(3, 3)	[1, 4], [2, 6], [3, 8], [5, 10], [7, 11], [9, 12]
(4, 3)	[1, 5], [2, 7], [3, 9], [4, 10], [6, 11], [8, 12]
(5, 3)	[1, 8], [2, 7], [3, 9], [4, 10], [5, 12], [6, 11]
(1, 4)	[1, 4], [2, 3], [5, 7], [6, 10], [8, 11], [9, 13], [12, 14], [15, 16]
(2, 4)	[1, 3], [2, 7], [4, 8], [5, 9], [6, 11], [10, 13], [12, 15], [14, 16]
(3, 4)	[1, 5], [2, 6], [3, 8], [4, 9], [7, 13], [10, 16], [11, 14], [12, 15]
(4, 4)	[1, 5], [2, 7], [3, 9], [4, 11], [6, 13], [8, 14], [10, 15], [12, 16]
(5, 4)	[1, 6], [2, 8], [3, 10], [4, 12], [5, 13], [7, 14], [9, 15], [11, 16]
(6, 4)	[1, 7], [2, 11], [3, 10], [4, 12], [5, 13], [6, 15], [8, 14], [9, 16]

7. MULTIPLICITY AT LEAST THREE

Proposition 7.1. *There is no three-fold Langford sequence of order 3 and defect 6, although $(m, d, l) = (3, 6, 3)$ satisfies both conditions of Theorem 3.1.*

Proof. The positions are $[1, 18]$ and the differences are 6, 7, 8, three pairs each. A position $x \leq 6$ is a left end, since $x - 6 < 1$; a position $x \geq 13$ is a right end, since $x + 6 > 18$. As in the proof of Theorem 3.1, $\sum R - \sum L = 3(6 + 7 + 8) = 63$ and $\sum R + \sum L = 171$, so $\sum L = 54$. The left ends are $[1, 6]$, with sum 21, and three positions of $[7, 12]$ with sum 33; since $10 + 11 + 12 = 33$ is the largest possible such sum, they are 10, 11, 12. Now 12 pairs only with $12 + 6 = 18$; 11 pairs with 17 or 18, hence with 17; 10 pairs with 16, 17 or 18,

hence with 16. These are three pairs of difference 6. But 7 is a right end, and its partner $7 - p$ with $p \geq 6$ is a left end in $[1, 6]$, so $7 - p = 1$ and $p = 6$: a fourth pair of difference 6. $\square$

Lean: `Langford.not_threeFold_six_three, L2.not_threeFold_six_three_internal (L2/SixThree.lean)`.

The conditions hold at $(3, 6, 3)$: $2 \cdot 6 + 3 = 15 \leq 19$ and $3(12 + 3 + 1) = 48 \equiv 0 \pmod{4}$.

Theorem 7.2 (order three). *For every $m \geq 3$ the cell $(d, l) = (3m - 3, 3)$ satisfies both conditions of Theorem 3.1, and no m -fold Langford sequence of order 3 and defect $3m - 3$ exists.*

Proof. Here $2d + l = 6m - 3 \leq 6m + 1 = 2ml + 1$, and for odd m , $l(2d + l + 1) = 18m - 6 = 6(3m - 1) \equiv 0 \pmod{4}$ because $3m - 1$ is even. The excess is $e = 3(2m - 1) - 2(3m - 3) + 1 = 4$ and $ml = 3m$, so Theorem 3.11 would give $12m \leq 36$, which fails for $m \geq 4$. At $m = 3$ the cell is $(6, 3)$, empty by Proposition 7.1. $\square$

Lean: `Langford.not_order_three, L2.not_order_three_internal (L2/Forced.lean)`.

The cells $(3m - 3, 3)$ lie two defects below the tight line of order three, which is $d = 3m - 1$. For $m = 2$ the cell is $(3, 3)$, which has a two-fold sequence (Theorem A; Table 3); so among the cells $(3m - 3, 3)$ with $m \geq 2$ exactly the one at $m = 2$ is realized.

Proposition 7.3 (the top of order two). *For every $m \geq 3$ there is no m -fold Langford sequence of order 2 and defect $2m - 1$.*

Proof. Here $ml = 2m$ and $e = 2(2m - 1) - 2(2m - 1) + 1 = 1$, so Theorem 3.11 would give $2m \leq 4$. (In the notation of §8, $F_1(2m - 1) = 2 - m < 0$.) $\square$

Lean: `L2.not_order_two_top_internal (L2/Forced.lean)`.

The defect $2m - 1$ is the largest the counting bound allows in order two ($2d + 2 \leq 4m + 1$). For even $m \geq 4$ the cell satisfies both conditions of Theorem 3.1, the parity condition being vacuous; for odd m the parity condition already excludes it ($l \equiv 2 \pmod{4}$), so it is a witness only for even m . At $m = 2$ the cell $(3, 2)$ is realized (Theorem A).

Theorem 7.4 (counting and parity do not suffice). *For every $m \geq 3$ there are $d, l \geq 1$ satisfying the conditions of Theorem 3.1 with no m -fold Langford sequence of order l and defect d .*

Proof. For $m = 3$ take $(d, l) = (6, 3)$ (Proposition 7.1). For even $m \geq 4$ take $(m - 1, 1)$: it is in-bound, parity is vacuous, and $m - 1 \nmid m$. For odd $m \geq 5$ take $(m - 2, 1)$: it is in-bound, $1 \cdot (2(m - 2) + 2) = 2m - 2 \equiv 0 \pmod{4}$, and $m - 2 \nmid m$ because $m - 2 \mid m$ would force $m - 2 \mid 2$. Theorem 3.6 shows the last two cells empty. $\square$

Lean: `Langford.not_sufficient, L2.not_sufficient_internal (L2/OrderOne.lean)`.

By Theorem 7.2 the witnesses may be taken uniformly in order three, $(d, l) = (3m - 3, 3)$ for every $m \geq 3$. Among other small witnesses, $(7, 4)$ at $m = 3$ is empty by Lemma 7.5 below and $(3, 1)$ at $m = 4$ by Theorem 3.6.

Lemma 7.5 (an emptiness lemma). *Let $m \geq 2$. If*

$$\frac{ml}{2} < d < \frac{m(2ml - 2l + 1)}{3m - 2},$$

then no m -fold Langford sequence of order l and defect d exists.

Proof. Put $N = 2ml$, $A = [1, d]$ and $C = [N - d + 1, N]$; these are disjoint since $d \leq ml$. Every position of A is a left end and every position of C a right end, and no pair lies inside A or inside C , since every difference is at least d . A pair from A to C has difference at least $N - 2d + 1$, and at most $t = \max(0, 3d + l - 1 - N)$ values $p \in [d, d + l - 1]$ are that large; so at most mt pairs join A to C , and at least $2d - 2mt$ positions of $A \cup C$ are paired with distinct positions of $B = [d + 1, N - d]$. Hence $N - 2d \geq 2d - 2mt$, that is $4d - N \leq 2mt$. If $t = 0$ this contradicts $d > ml/2$, which says $4d - N > 0$. If $t > 0$ it contradicts $d < m(2ml - 2l + 1)/(3m - 2)$, which after expanding says exactly $4d - N > 2m(3d + l - 1 - N) = 2mt$. $\square$

Status: proved here; not formalized.

The interval of Lemma 7.5 has length $ml(m - 2)/(2(3m - 2)) + m/(3m - 2)$: less than one at $m = 2$, where it contains no integer (as Theorem A requires), $3l/14 + 3/7$ at $m = 3$, $2l/5 + 2/5$ at $m = 4$, and about $ml/6$ only as m grows.

Computations for $m \geq 3$. Every in-bound cell has been decided for $m = 3$ with $l \leq 24$, $m = 4$ with $l \leq 20$, and $m = 5, 6$ with $l \leq 12$, each by a checked sequence, an exact integer Farkas certificate (§8), or the parity condition. In every row there is an empty band of cells satisfying both conditions of Theorem 3.1: at $m = 3$ it is $d \in [11, 18]$ at $l = 8$, $[15, 28]$ at $l = 12$ and $[30, 55]$ at $l = 24$; at $m = 4$, $[14, 27]$ at $l = 8$ and $[34, 67]$ at $l = 20$. The last realized d below the band is about $1.2l$, $1.64l$, $2.1l$ and $2.6l$ for $m = 3, 4, 5, 6$.

Status: checked on the stated ranges; every witness and every certificate is verified by a standard-library checker with exact arithmetic; not formalized.

8. THE LINEAR-PROGRAMMING PICTURE FOR $m \geq 3$

Nothing in this section is formalized; it is proved here, and its proofs and finite checks were re-derived independently. Fix (m, d, l) and $N = 2ml$.

Lemma 8.1 (Farkas certificates). *Let $y_1, \dots, y_N$ and $z_d, \dots, z_{d+l-1}$ be integers with $y_a + y_{a+p} + z_p \geq 0$ whenever $1 \leq a < a + p \leq N$ and $p \in [d, d + l - 1]$. If $\sum_x y_x + m \sum_p z_p < 0$, the cell has no m -fold sequence, and its linear relaxation (fractional pairings) is infeasible.*

Proof. Sum the inequality over the ml pairs of a sequence, or over a fractional pairing with its weights: every position is covered once and every p exactly m times. $\square$

The counting bound is the certificate $y_x = |2x - N - 1|$, $z_p = -2p$, whose total is $ml((2m - 1)l - 2d + 1)$.

Lemma 8.2 (a family of certificates). *For an integer $T \geq 1$ and a sign $s = \pm 1$, put $c = (N + 1)/2$, $W_T(u) = T/2 - \text{dist}(u, 2T\mathbb{Z})$, $g(x) = 2sW_T(x - c)$ and $z_p = 2|p - T|$. Then g is integer-valued and $y = g$, z satisfy the inequalities of Lemma 8.1. Moreover*

$$\Phi_s(T) = \sum_x g(x) + m \sum_p z_p = 2s\sigma_2(ml \bmod 2T) + 2m \sum_{p=d}^{d+l-1} |p - T|,$$

where $\sigma_2(r) = r(T - r)$ for $0 \leq r \leq T$ and $(r - T)(r - 2T)$ for $T \leq r \leq 2T$. The counting bound is the member $T \geq \max(ml, d + l - 1)$, $s = -1$, with $\Phi = ml((2m - 1)l - 2d + 1)$.

Proof. $2W_T(x - c) = T - \text{dist}(2x - N - 1, 4T\mathbb{Z})$ is an integer. W_T is 1-Lipschitz and $W_T(u + T) = -W_T(u)$, so with $u = a - c$, $g(a) + g(a + p) = 2s[W_T(u) - W_T(u + p - T)] \geq -2|p - T|$. For the closed form, g is symmetric about c , so $\sum g = 4s \sum_{k=1}^{ml} W_T(k - \frac{1}{2})$; the sum over a period $2T$ vanishes and the partial sums are $r(T - r)/2$ and $(r - T)(r - 2T)/2$. $\square$

Since $|\sigma_2(ml \bmod 2T)| = r(T - r)$ with $r = ml \bmod T$, the smaller of the two totals is $\min_s \Phi_s(T) = 2H(T)$, where

$$H(T) = mS(T) - r(T - r), \quad S(T) = \sum_{p=d}^{d+l-1} |p - T|;$$

$H(T)$ is half the minimum certificate total, and the statement that it is nonnegative whenever a sequence exists is the residue bound, Theorem 3.7, which is formalized. With Theorem A it gives the inequality of Proposition 8.6 below, though not its equality cases.

Theorem 8.3 (explicit empty bands). *Let $m \geq 3$, $j \geq 1$ with $J = j(j + 1) < m$, and $l \geq 2$. Suppose d is an integer with*

$$|2Jd - ((2j + 1)m - J)l - J| < \sqrt{(m^2 - J^2)l^2 - J^2(J + m)/m},$$

let T_0 be an integer nearest to $T^ = m((2j + 2)l + 2d - 1)/(2(J + m))$, and suppose $d \leq T_0 \leq d + l - 1$ and $jT_0 \leq ml \leq (j + 1)T_0$. Then the cell (m, d, l) is empty, certified by Lemma 8.2 with $T = T_0$ and $s = (-1)^{j+1}$. For $j = 1$ the side conditions on T_0 follow from $2d \geq (m - 2)l + m + 3$.*

Proof outline. In the regime $jT \leq ml \leq (j + 1)T$, $d \leq T \leq d + l - 1$ one has $\Phi = 2F_j(T)$ with $F_j(T) = (J + m)T^2 - m((2j + 2)l + 2d - 1)T + m^2l^2 + md^2 + m(l - 1)d + m(l^2 - l)/2$. Rounding T^* to T_0 costs at most $(J + m)/4$. Writing $4(J + m)C - b^2 = mQ_j(d)$ for the constant C and the linear coefficient b of F_j , with $Q_j(d) = 4Jd^2 - 4((2j + 1)ml - Jl + J)d + 4m^2l^2 - (4j + 2)ml^2 + 2Jl^2 + (4j + 2)ml - 2Jl - m$, whose discriminant is $16[(m^2 - J^2)l^2 + J(J + m)]$, the hypothesis is equivalent to $mQ_j(d) + (J + m)^2 < 0$, whence $F_j(T_0) < 0$. $\square$

Status: proved here (every algebraic identity also confirmed symbolically); checked exactly on $3 \leq m \leq 30$, $2 \leq l \leq 200$; for $j \geq 2$ the regime conditions are hypotheses, observed to hold at every tested cell; not formalized.

Lemma 8.4 (the outer bound). *In the regime $jT \leq ml \leq (j + 1)T$, $d\Phi/dT = 4(j + 1)(jT - ml) \leq 0$ for $T < d$ and $d\Phi/dT = 4j((j + 1)T - ml) \geq 0$ for $T > d + l - 1$. So within a regime the members with $T \in [d, d + l - 1]$, where $\Phi = 2F_j(T)$, dominate the others whenever that window meets the regime; and if such a member certifies the cell, then $\min_{T \in \mathbb{R}} F_j(T) < 0$, that is $Q_j(d) < 0$: d lies strictly between the outer roots $[(2j + 1)ml - Jl + J \mp \sqrt{(m^2 - J^2)l^2 + J(J + m)}]/(2J)$.*

Status: proved here; not formalized. Every one of the 14,975 cells certified by the family in the ranges scanned lies strictly inside the outer interval of some j with positive discriminant (checked).

Over $m \leq 30$, $l \leq 200$ the outer radius exceeds the inner radius of Theorem 8.3 by at most 0.725. Letting $l \rightarrow \infty$:

Corollary 8.5 (band edges for large l). *For each $j \geq 1$ with $J = j(j + 1) < m$, the family of Lemma 8.2 certifies an empty band in row l whose edges satisfy*

$$\frac{d}{l} \longrightarrow \frac{(2j + 1)m - J \mp \sqrt{m^2 - J^2}}{2J} \quad (l \rightarrow \infty).$$

Status: proved here from Theorem 8.3 and Lemma 8.4, as a statement for large l ; not formalized.

Table 4 gives the limits for small m . The first band ends below the tight line at excess $e/l \rightarrow (m - \sqrt{m^2 - 4})/2$, where $e = (2m - 1)l - 2d + 1$; this is 0.382, 0.268, 0.209, 0.172 for $m = 3, \dots, 6$. The second band ($j = 2$) exists from $m = 7$ on and the third from $m = 13$ on.

TABLE 4. Limits of d/l at the edges of the empty band j (Corollary 8.5).

m	$j = 1$	$j = 2$	$j = 3$
3	(1.191, 2.309)	—	—
4	(1.634, 3.366)	—	—
5	(2.104, 4.396)	—	—
6	(2.586, 5.414)	—	—
7	(3.073, 6.427)	(2.116, 2.717)	—
8	(3.564, 7.436)	(2.392, 3.274)	—
13		(3.956, 5.878)	(3.083, 3.500)

The gap between consecutive bands grows linearly in l , roughly as $(j + 1)^2 l / (2m)$ for large m . At small l further bands appear: the outer interval of a j with $J > m$ is nonempty exactly when $l^2 < J/(J - m)$, which accounts for eight certified cells in the row $l = 1$ outside every $J < m$ interval, such as $(m, d, l) = (5, 2, 1)$. At $m = J$ ($m = 2, 6, 12, 20$) a band touches zero at $d \in \{jl, jl + 1\}$, with $\min \Phi = 0$ (checked for $l \leq 30$).

On every row decided in §7 and on three further complete rows, the band certified by this family equals the set of cells certified by general Farkas certificates (Lemma 8.1): 1,219 cells, none of them realized. All 2,840 emptiness certificates produced in the computations were re-verified by independent code, and every certified cell with $N \leq 30$ was confirmed empty by exhaustive search.

Proposition 8.6 (no obstruction of this kind at $m = 2$). *For $m = 2$, every in-bound (d, l) , every integer $T \geq 1$ and both signs, $\Phi_s(T) \geq 0$. Equality holds only for $s = +1$ with $d = l$, $2T - 3l \in \{-1, 0\}$ or $d = l + 1$, $2T - 3l \in \{0, 1\}$; for $s = -1$ on the tight line with $T \geq \max(2l, d + l - 1)$; and in two degenerate cases ($T = 1$ at $(d, l) = (1, 1)$, and $(d, l, T, s) = (2, 1, 2, +1)$). So within this family the counting bound is the only member that ever refutes a cell at $m = 2$.*

Proof outline. With $ml = 2l$: for $T \geq 2l$, $\sigma_2 = 2l(T - 2l) \geq 0$, so $s = +1$ is harmless, and for $s = -1$, $|p - T| \geq T - p$ gives $\Phi \geq 2l(3l - 2d + 1) \geq 0$. For $l \leq T < 2l$, $\sigma_2 \leq 0$ makes $s = -1$ harmless for $l \geq 2$, and for $s = +1$, $T = l + t$, $\Phi = 4[\sum_p |p - T| - (l - t)t] \geq 0$ because the distance sum from an integer to l consecutive integers is at least $\lfloor l^2/4 \rfloor$. For $T < l$, $|\sigma_2| \leq T^2/4$ and $\Phi > 0$. $\square$

Status: proved here; checked exactly for $l \leq 80$, $T \leq 8l$, both signs, by two implementations; not formalized. It concerns this centred, integer- T family only.

Lemma 8.7 (blow-up). *If an m -fold Langford sequence of order l_0 and defect d_0 exists, then for every odd k so does one of order kl_0 and defect $kd_0 - (k - 1)/2$.*

Proof. Replace position a by the block $[k(a - 1) + 1, ka]$, and each pair (a, b) by the k pairs $(k(a - 1) + i, k(b - 1) + \tau_k(i))$, $i \in [1, k]$. The displacements of τ_k are $[-(k - 1)/2, (k - 1)/2]$

once each, so a difference p becomes the k differences $kp + t$, $|t| \leq (k-1)/2$, and consecutive p tile $[kd_0 - (k-1)/2, kd_0 - (k-1)/2 + kl_0 - 1]$, each value m times. $\square$

Status: proved here; checked on 264 instances; not formalized. The same holds for cells with feasible relaxation. We have not checked the literature for this multiplication.

The quadratic inequality without regime hypotheses. For $j \geq 1$ put

$$J = j(j+1), \quad a = m + J, \quad b = m((2j+2)l + 2d - 1),$$

$$C = m^2l^2 + md^2 + m(l-1)d + \frac{ml(l-1)}{2},$$

and $F_j(T) = aT^2 - bT + C$. The coefficients are integers, and F_j is the polynomial of Theorem 8.3.

Theorem 8.8 (the quadratic inequality). *If an m -fold Langford sequence of order l and defect d exists, then $F_j(T) \geq 0$ for every $j \geq 1$ and every integer T .*

Proof. Put $n = ml$. For an integer $T \geq 1$ let

$$K_j(T) = (n - jT)(n - (j+1)T), \quad P(T) = T^2 - (2d + l - 1)T + d^2 + (l-1)d + \frac{l(l-1)}{2};$$

expanding, $F_j(T) = K_j(T) + mP(T)$. Two inequalities hold for every integer $T \geq 1$, whether or not a sequence exists.

First, write $n = qT + r$ with $0 \leq r < T$ and put $h = j - q$. Then

$$K_j(T) + r(T - r) = hT((h+1)T - 2r),$$

which is positive for $h \geq 1$, zero for $h = 0$, and for $h \leq -1$ a product of two factors that are at most 0. So

$$(3) \quad K_j(T) \geq -r(T - r).$$

Second, $P(T) \geq S(T)$. If $d \leq T \leq d + l - 1$ and $u = T - d$, then $P(T) = u^2 - (l-1)u + l(l-1)/2$, which is $S(T)$ by the closed form in the proof of Corollary 3.9. If $T < d$, then $P(T) - S(T) = u(u-1)$ with $u = d - T \geq 1$; if $T > d + l - 1$, then $P(T) - S(T) = v(v-1)$ with $v = T - (d + l - 1) \geq 1$. Both are nonnegative because u and v are positive integers.

Hence $F_j(T) \geq mS(T) - r(T - r) = H(T)$ for every integer $T \geq 1$, and $H(T) \geq 0$ by Theorem 3.7. For $T \leq 0$ the three terms aT^2 , $-bT$ and C are nonnegative and $C > 0$. $\square$

Status: proved here; not formalized.

Theorem 8.3 is the special case $T = T_0$ with its regime hypotheses, which make $\Phi = 2F_j(T_0)$, and with the worst-case rounding cost $a/4$. Theorem 8.8 needs neither; in particular the regime conditions, which Theorem 8.3 takes as hypotheses for $j \geq 2$, are not needed for the emptiness of the cell. The forced-endpoint bound is $2F_1(d) \geq 0$ (Theorem 3.11). A negative $F_j(T)$ at an integer $T \geq 1$ is a negative $H(T)$, that is, a negative member of the family of Lemma 8.2 at the same T .

Exact rounding and a finite test. Put

$$t_j = \left\lfloor \frac{b+a}{2a} \right\rfloor, \quad \rho_j = b - 2at_j,$$

so that t_j is an integer nearest to $b/(2a)$, ties rounded up, and put

$$X = 2Jd - ((2j+1)m - J)l - J, \quad D = (m^2 - J^2)l^2 + J(m+J).$$

Proposition 8.9 (exact rounding and a finite cutoff). *Let $j \geq 1$, with t_j , ρ_j , X and D as above.*

- (i) $-a \leq \rho_j < a$, t_j minimizes F_j over the integers, and $4aF_j(t_j) = 4aC - b^2 + \rho_j^2$. So $F_j(t_j) < 0$ if and only if $4aC - b^2 + \rho_j^2 < 0$.
- (ii) $J(4aC - b^2) = m(X^2 - D)$ and $D = (m+J)(ml^2 - J(l^2 - 1))$. So $F_j(t_j) < 0$ if and only if $mX^2 + J\rho_j^2 < mD$.
- (iii) If $F_j(t_j) < 0$, then $J(l^2 - 1) < ml^2$.

Proof. (i) The bounds on ρ_j restate $t_j \leq (b+a)/(2a) < t_j + 1$. For every T , $4aF_j(T) = (2aT - b)^2 + 4aC - b^2$, so F_j is smallest over the integers at an integer nearest to $b/(2a)$, and $2at_j - b = -\rho_j$. (ii) Both identities are checked by expanding. (iii) If $F_j(t_j) < 0$, then $4aC - b^2 < -\rho_j^2 \leq 0$, so by (ii) $D > X^2 \geq 0$, and $m+J > 0$. $\square$

Status: proved here; checked exactly on the grid described below; not formalized.

Part (i) is a test in exact integer arithmetic, with no square root. Theorem 8.3 replaces ρ_j^2 by its worst case a^2 : its hypothesis is $mX^2 + Ja^2 < mD$, and the outer interval of Lemma 8.4 is $X^2 < D$. For $l \geq 2$, part (iii) gives $J < ml^2/(l^2 - 1) \leq 4m/3$, so only the indices with $j(j+1) < 4m/3$, of which there are $O(\sqrt{m})$, need to be tested; this counts tests, each an exact comparison of integers, not bit operations. The cutoff cannot be simplified to $J < m$, the range of Theorem 8.3: Example 8.12 below has $J = 20 > m = 19$. For $l = 1$ the condition in (iii) is vacuous; that row is settled by Theorem 3.6.

Theorem 8.10 (completeness for the family). *Let $l \geq 2$. Some member of the family of Lemma 8.2, an integer $T \geq 1$ with a sign s , has $\Phi_s(T) < 0$ if and only if the counting bound fails, $(2m-1)l < 2d-1$, or $F_j(t_j) < 0$ for some $j \geq 1$ with $J(l^2 - 1) < ml^2$.*

Proof. Since $\min_s \Phi_s(T) = 2H(T)$, the left side says that $H(T) < 0$ for some integer $T \geq 1$. Put $n = ml$ and $e = (2m-1)l - 2d + 1$.

If the counting bound fails, the member $T \geq \max(n, d+l-1)$, $s = -1$, has $\Phi = ne < 0$. If $F_j(t_j) < 0$, then $t_j \geq 1$, because $F_j > 0$ at the integers $T \leq 0$, and $H(t_j) \leq F_j(t_j) < 0$ by the proof of Theorem 8.8.

Conversely, suppose that the counting bound holds and $H(T) < 0$. By Proposition 8.9(iii) it suffices to find $j \geq 1$ and an integer T' with $F_j(T') < 0$, since then $F_j(t_j) \leq F_j(T') < 0$. If $T \geq n$, then $mS(T) \geq m \sum_p (T-p) = nT - m \sum_p p$ and $r(T-r) = n(T-n)$, so $H(T) \geq n^2 - m \sum_p p = ne/2 \geq 0$, a contradiction. So $T < n$; put $j = \lfloor n/T \rfloor \geq 1$. Then $jT \leq n < (j+1)T$, so $h = 0$ in the proof of Theorem 8.8, and $H(T) = K_j(T) + mS(T)$.

- If $d \leq T \leq d+l-1$, then $S(T) = P(T)$ and $F_j(T) = H(T) < 0$.
- If $T < d$, let $\eta(u) = K_j(u) + m \sum_p (p-u)$ for real u . Then $\eta(T) = H(T)$, and $\eta'(u) = 2(j+1)(ju-n) \leq 0$ for $u \leq n/j$. If $n/j < d$, then $\eta(n/j) \leq \eta(T) < 0$; but $K_j(n/j) = 0$ and every p exceeds n/j , so $\eta(n/j) > 0$. Hence $T < d \leq n/j$, and $F_j(d) = \eta(d) \leq \eta(T) < 0$, since $\sum_p (p-d) = S(d) = P(d)$.

- If $T > d + l - 1$, let $\eta(u) = K_j(u) + m \sum_p (u - p)$ for real u . Then $\eta(T) = H(T)$, and $\eta'(u) = 2j((j+1)u - n) \geq 0$ for $u \geq n/(j+1)$. If $n/(j+1) > d + l - 1$, then $\eta(n/(j+1)) \leq \eta(T) < 0$; but $K_j(n/(j+1)) = 0$ and every p is below $n/(j+1)$, so $\eta(n/(j+1)) > 0$. Hence $n/(j+1) \leq d + l - 1 < T$, and $F_j(d + l - 1) = \eta(d + l - 1) \leq \eta(T) < 0$. $\square$

Status: proved here; checked exactly on the grid described below; not formalized.

The real variable u is used only on intervals where the distance sum is linear in u ; the closed form of S inside the window is not used at non-integer points. The theorem is completeness for this family only: the finite test detects exactly the negative members of the family of Lemma 8.2. It is not completeness for the linear relaxation, whose infeasibility may have other certificates, and it says nothing about existence at cells that pass the test; it proves neither Conjecture 8.13 below nor the stronger guess stated after it. For $l = 1$ the family is exact by Theorem 3.6: at $T = d$, $S(d) = 0$ and $H(d) = -r(d - r)$ with $r = m \bmod d$, which is negative exactly when $d \nmid m$, and when $d \mid m$ a sequence exists, so by Lemma 8.1 no member of the family is negative.

Example 8.11. The cell $(m, d, l) = (7, 9, 4)$ satisfies the counting bound ($22 \leq 57$) and the parity condition ($4 \cdot 23 \equiv 0 \pmod{4}$). For $j = 2$, $(a, b, C, t_2, \rho_2) = (13, 287, 1582, 11, 1)$ and $F_2(11) = -2$, so the cell is empty. The worst-case expression of Theorem 8.3 is $4aC - b^2 + a^2 = 64 > 0$, so its inner test does not reach this cell, while $4aC - b^2 + \rho_2^2 = -104$. In the form of Theorem 3.7, at $T = 11$: $ml \bmod 11 = 6$ and $7(2+1+0+1) = 28 < 6 \cdot (11-6) = 30$.

Example 8.12. The cell $(m, d, l) = (19, 12, 3)$ satisfies the counting bound and the parity condition ($3 \cdot 28 \equiv 0 \pmod{4}$). For $j = 4$, so $J = 20 > m$, $(a, b, C, t_4, \rho_4) = (39, 1007, 6498, 13, -7)$ and $F_4(13) = -2$, so the cell is empty. The index lies inside the cutoff, $J(l^2 - 1) = 160 < ml^2 = 171$, and outside the range $J < m$ of Theorem 8.3. In the form of Theorem 3.7, at $T = 13$: $57 \bmod 13 = 5$ and $19(1+0+1) = 38 < 5 \cdot 8 = 40$.

Status: exact integer arithmetic, repeated by the checker described below; not formalized.

The two uniform families of §7 in this language: at $(d, l) = (3m - 3, 3)$, $F_1(3m - 2) = -4(m - 2)$, negative for every $m \geq 3$, including $m = 3$; at $(d, l) = (2m - 1, 2)$, $F_1(2m - 1) = 2 - m$. In both, T lies in the window and $\lfloor ml/T \rfloor = 1$, so these values are $H(T)$.

Finite checks. A standard-library checker in exact integer arithmetic tested this subsection on the grid $1 \leq m \leq 20$, $1 \leq l \leq 20$, $1 \leq d \leq \lfloor ((2m - 1)l + 1)/2 \rfloor + 2$, which runs two defects past the counting line: 42,900 triples. On every triple the finite test (the counting bound, then $F_j(t_j)$ for the indices within the cutoff) agrees with a direct scan of $H(T)$ over every integer T with $1 \leq T \leq \max(ml, d + l - 1)$, beyond which H is constant. The test rejects 28,645 triples that satisfy the counting bound; for the 1,026 of them with $m \leq 10$ and $l \leq 8$ the checker rebuilt the integer weights of the certificate, checked every edge inequality and confirmed the negative total. On the grid, 98 first-detected obstructions are missed by the worst-case rounding inequality of Theorem 8.3; this count includes cells of order one and cells that fail the parity condition, and is not a count of new empty cells. The checker also verified the two examples, the identities $4d^2 - 6mld + 2m^2l^2 + ml(l - 1) = 2F_1(d) = (l - 1 + e)^2 - mle$ for $m, l \leq 30$ and $d \leq ml$, and the identities of the two uniform families for $3 \leq m \leq 1000$, and it reproduces the empty bands of §7 in the rows $(m, l) = (3, 8)$, $(3, 12)$, $(3, 24)$ and $(4, 20)$: $d \in [11, 18]$, $[15, 28]$, $[30, 55]$ and $[34, 67]$. A second check decided every cell with $2ml \leq 20$ and $m \leq 4$, up to one defect past the counting line, by exhaustive search (79 cells, 41 realized): the residue bound, for every $T \leq 2ml + 2$, and the forced-endpoint bound

hold at every realized cell, the equality case of Theorem 3.1, in its sequence form, holds for every sequence found, and among the 38 empty cells the residue bound rejects 28 and the forced-endpoint bound 8. At $m = 2$, where Theorem A realizes every in-bound cell, both bounds hold at every in-bound cell with $l \leq 400$, the residue bound for every $T \leq 4l + 2$; the second check also confirmed the two examples.

Realized cells between empty bands. For $m \geq 7$ a row can contain two empty bands with realized cells between them. In the row $m = 7$, $l = 12$ the cells $d = 33$ and $d = 38$ are empty by exact certificates while $d = 34, \dots, 37$ are realized. Such islands are witnessed, each by a checked sequence flanked by certified empty cells in its row, for $m \in \{7, 8, 9, 11, 13, 14, 15\}$ (for instance $(m, d, l) = (8, 34, 10), (9, 12, 3), (11, 20, 4), (13, 15, 4), (14, 21, 5), (15, 18, 4)$); and for every even $m \geq 8$ by Theorem 3.6 in the row $l = 1$, where $d = m/2$ is realized while $d = m/2 - 1$ and $d = m/2 + 1$ are not. For other odd m they are not yet witnessed. For $m \leq 6$ the certified band is a single interval in every row checked ($l \leq 60$).

Conjecture 8.13 ($C(m)$). *For every $m \geq 3$, an m -fold Langford sequence of order l and defect d exists if and only if the linear relaxation of the cell is feasible and, for odd m , the parity condition holds.*

It holds on every decided cell of §7 and on the islands above. A stronger guess is that the certificates of Lemma 8.2 are the only binding ones, so that the relaxation is infeasible exactly when some $\Phi_s(T) < 0$; they agree on all 1,219 band cells of the decided rows and, in floating point, on 635 further cells with $m \leq 14$. With Proposition 8.6, that guess would make the relaxation feasible at every in-bound cell for $m = 2$, which Theorem A implies anyway. No closed form for the edges of the true empty set is known, and $C(m)$ is open.

9. THE FINITE DATA AND THE FORMAL DEVELOPMENT

The end-to-end certificate. The certificate folder under `note/certificates/` of the repository [15] holds two standard-library Python checkers, `verify.py` and `verify_second.py`. Each rebuilds, for every $1 \leq l \leq L$ and every $1 \leq d \leq \lfloor (3l + 1)/2 \rfloor$, a two-fold sequence by the constructions of this note alone — concatenation for splittable cells; for the rest, Lemma S applied to a permutation obtained by the reversal, τ_l , the descent of Lemma 4.9, or a band family, with the bases and sporadic cells — and checks it from the definition (the positions partition $[1, 4l]$; each difference in $[d, d + l - 1]$ occurs exactly twice). The two scripts were written independently, share no code, and transcribe every family table separately. The first also checks each permutation in displacement form before lifting, runs every family over its whole domain with 63 mutated families rejected, and re-finds the eight sporadic permutations by its own exact search; the second checks the domain of every family to $l = 160$, compares its closed-form k -step descent with the iterated one on 2,701 chains, and rejects 16 mutated sequences. Both also check the m -fold tight line for $m \leq 6$ and odd $l \leq 41$.

The first script ran to $L = 1000$ (751,000 cells) and the second to $L = 600$, with no failure; both reproduce at $L = 300$ in fifteen to twenty seconds with `python verify.py 300` and `python verify_second.py 300`. At $L = 300$ the 67,800 in-bound cells are reached by 25,331 concatenations, 600 reversals, 150 applications of τ_l , 19,665 band-family cells, 22,052 descent chains and 2 literal sequences (the cells $(d, l) = (3, 4)$ and $(6, 4)$, where SP is empty); the route counts of the two scripts agree class by class up to the choice between a literal and a construction at $l \leq 4$ and the routing of $(7, 3)$.

The finite data establish nothing about all l : the certificate is the finite witness of the theorem, checked to $l = 1000$ by one implementation and to $l = 600$ by a second, while the statement for all l rests on the class tables, Theorem R and the induction, and is the formalized Theorem A.

Other computations. Independently of the constructions, every in-bound cell at $m = 2$ with $l \leq 54$ was realized by a constraint solver or by concatenation, and the cells with $l \leq 14$ by a second, exhaustive engine; and the numbers of two-fold Skolem sequences ($d = 1$) of orders $1, \dots, 8$ were recomputed as 1, 3, 12, 186, 3212, 79238, 2770026, 127860956, in agreement with Grüttmüller, Rees and Shalaby [6] as tabulated in [3]. Before the band families were found, the signed-permutation problem was solved by exact cover at every cell with $l \leq 100$ except the two empty cells of Lemma 4.6.

The formal development. The repository [15] pins Lean v4.35.0-rc2 and the Mathlib tag of the same name by a committed manifest. `Challenge.lean` states the definition and the ten theorems of §1; `Solution.lean` restates them byte for byte and closes each with the internal theorem of the same name in the library L2, which has these modules:

- **Defs** — the verbatim copy of `Langford.IsLangford`;
- **Pairs** — endpoints, differences and translations of finite sets of pairs, ported from [14];
- **Rows** — the row of a block reversal, its values and mirrors (Lemma 5.1);
- **SP** — SP in graph form, inversion, the reversal, τ_l , Lemma C and the descent (§4);
- **FamROa, FamROb, FamR1, FamR2, FamR3** — one definition and one theorem per family, each the class table of Appendix A with the audited domain as hypotheses;
- **Literals** — the eleven permutations of Table 2 and the sixteen cells of Table 3;
- **Band, Cone** — Theorems 5.3–5.7 and Theorem 4.11;
- **Multi, LemmaS, Bridge** — certificates, concatenation, juxtaposition, Lemma S and Lemma 2.5;
- **Tight, Necessity, OrderOne, SixThree** — Theorems 3.3, 3.1, 3.6, 7.4 and Proposition 7.1;
- **Residue** — the residue bound, Theorem 3.7 and Corollary 3.9;
- **Forced** — the forced-endpoint bound, the order-three cells and the top of order two, Theorems 3.11 and 7.2 and Proposition 7.3;
- **Straddle** — the rigidity of the counting bound, the equality clause of Theorem 3.1;
- **Main** — the induction of §6 and the internal theorems.

Internals work over $\mathbb{Z}$, where linear arithmetic with negative slopes is decided by Lean’s `omega`; the pinned statements are over $\mathbb{N}$, and the casts happen in `Bridge` and `Main`, and inside the proof of the residue bound, whose potential takes negative values. Finite facts — the eleven permutations, the sixteen literal cells, and in the proof for (6,3) the three-element subsets of $[7, 12]$ with sum 33 — are decided by the kernel on literal finite sets; there is no other search inside the proofs. A build-time audit (`Test/Axioms.lean`) walks every declaration of the development — `DESK_FILLS` constants in about `DESK_FILLS` lines of Lean — and admits only the axioms `propext`, `Classical.choice` and `Quot.sound`; a source guard (`scripts/check-source.py`) rejects `sorry`, `axiom`, `native_decide`, `implemented_by`, `extern` and the kernel-bypass options. A standard-library script, `scripts/check_langford.py`, recomputes the finite content independently of Lean: the sixteen literal cells and the eleven permutations from the definitions, every family at every domain cell with $l \leq 120$ lifted through Lemma S, the m -fold tight line for $m \leq 6$ and

odd $l \leq 21$, the row $l = 1$ for $m \leq 12$, the emptiness of $(3, 6, 3)$ by exhaustive search, the bounds of Theorems 3.7 and 3.11 against brute-force existence on small cells, and two controls that must fail.

10. PRIOR ART

On 2026-09-26 searches were made of the full texts of [1] (arXiv version and journal version) and of the thesis [2]; of seven further theses from the same department; of Nordh's papers on perfect Skolem sets; of arXiv, Crossref, zbMATH Open, the On-Line Encyclopedia of Integer Sequences, a public repository of formalized conjectures, and the Palomar registry of machine-checked results; and of the works citing [1], [4] and Nordh's papers, and the bibliography of the survey [5]. No statement of necessary and sufficient conditions for two-fold or m -fold Langford sequences with $d \geq 2$ was located. The question is printed as open in [1] (§7, identical in both versions) and twice in [2]; Table 8 of [1], which summarizes known existence results, has m -fold Skolem rows and no m -fold Langford row. The search was bounded, and on 2026-09-27 its blind spots were re-examined. The body of the survey [5] and the Handbook chapter on Skolem-type sequences could be searched only through full-text indexes, which contain no m -fold Langford statement; the 2026 paper of Zhang and Cao on generalized signed Langford sequences (its abstract and references) and the 2017 paper of López and Muntaner-Batlle on distance labelings (read in full) concern different objects — a signed Langford sequence is, by Nordh's Theorem 15, a one-fold sequence with a different difference set; Nowakowski's 1975 master's thesis and Dillon's 1973 paper on the generalized Langford–Skolem problem remain unread, and MathSciNet was not searched. Failure to retrieve a paper is not evidence about its contents, and nothing here is a claim of priority beyond that wording.

The residue bound and the forced-endpoint bound were the subject of a targeted search on 2026-09-27 (the same sources, with the terms weight function, linear programming and Farkas); no condition of this kind was located, and Nordh's parity and density conditions (Remark 3.2) are the only comparable ones, reducing to the counting bound. No claim of novelty is made beyond that wording.

Four connections follow.

Perfect Skolem sets. As Remark 3.2 explains, the objects of this note are Nordh's perfect multisets $m \times [d, d + l - 1]$ [10]; his Theorem 15 is the tight-line bijection, and his necessary conditions reduce to Theorem 3.1. Nordh's introduction notes that the existence question for general multisets is NP-complete; the multisets here are highly structured, and that result is context only.

Constructions in [1]. Construction 2.7 there is a two-fold Langford family on the tight line, and the sequences obtained by concatenating two Langford sequences, or interlacing a hooked one with its reverse, exist exactly when $l \geq 2d - 1$ or $(d, l) \in \{(2, 1), (2, 2)\}$. Nothing in print constructs two-fold Langford sequences inside the unsplittable region.

Chains and multiplicity. The generalized Langford sequences of multiplicity s (the copies of each value in one chain with equal gaps; see [9]) are a different object. A chain of four copies contains two disjoint pairs, so such sequences with $s = 4$ are two-fold Langford sequences, but their span forces $l \geq 3d - 2$, far from the unsplittable region.

Defect one. The m -fold Skolem sequences of [4], and their descendants (near, hooked, extended and disjoint variants), are all of defect 1. The exact counts of two-fold Skolem sequences of [6] are reproduced by the search (§9).

In the registry of machine-checked results (read to 2026-09-27), Langford sequences appear only as sources of the formalization [14] and of its successor; the repository of formalized conjectures contains no Langford or Skolem statement.

PROVENANCE

This note and the accompanying Lean development were produced with Claude (Anthropic), running with parallel Opus subagents; the mathematics, the searches and the checkers arose in that process. JD Jones owns and directed the project and is responsible for the submission. No independent human review or endorsement by the cited authors is claimed.

APPENDIX A. THE CLASS TABLES

Conventions (§5). Sizes, $M_i = 2(c_i - a_i) + \mu$ and the class runs are affine in (l, μ) on the family's residue class; for the line families they are written in l alone. The target order lists the block indices from left to right in target position. The type is P, N, C1 or C3 as in Lemma 5.1; a C block contributes two runs, b_i^+ (its positive x) and b_i^- (its negative x). The two chains list the runs of each parity in increasing order; in each chain the first run starts at 1 or 2, each run starts 2 above the end of the previous one, and the last ends at l or $l - 1$, all identically in (l, μ) . Every domain below is the set of cells of the residue class with μ odd at which every size is at least 1 and every type condition holds. The tables were generated from the sizes and orders of the audited transcription and every identity was checked symbolically; every family was also built at every domain cell with $l \leq 240$ and checked in displacement form.

FA. $l \equiv 2 \pmod{4}$; domain $(l + 4)/2 \leq \mu \leq (3l - 2)/4$. Target order $(2, 3, 0, 4, 1)$. Lean: `famFA`.

block	n_i	M_i	type	class runs
b_0	$(\mu - 1)/2$	$2l - \mu$	P	odd $[l - \mu + 2, l - 1]$
b_1	$(l + 2)/4$	$3l/2$	P	even $[(l + 2)/2, l]$
b_2	$(3l - 4\mu + 2)/4$	$-l/2$	N	odd $[(2\mu - l + 2)/2, l - \mu]$
b_3	$(l - 2)/4$	$-l/2$	N	even $[2, (l - 2)/2]$
b_4	$(2\mu - l)/4$	$(2\mu - l - 2)/2$	P	odd $[1, (2\mu - l - 2)/2]$

Odd chain: b_4, b_2, b_0 . Even chain: b_3, b_1 .

FB2. $l \equiv 2 \pmod{8}$; domain $(3l + 6)/4 \leq \mu \leq l - 1$ (so $l \geq 10$). Target order $(1, 3, 0, 4, 2)$. Lean: `famFB2`.

block	n_i	M_i	type	class runs
b_0	$(4\mu - l - 2)/8$	$(9l - 4\mu - 2)/4$	P	odd $[(5l - 4\mu + 6)/4, l - 1]$
b_1	$(l - 2)/8$	$(l + 2)/4$	P	even $[2, (l - 2)/4]$
b_2	$(3l + 2)/8$	$(5l + 2)/4$	P	even $[(l + 6)/4, l]$
b_3	$l - \mu$	$-l/2$	N	odd $[(4\mu - 3l + 6)/4, (5l - 4\mu - 2)/4]$
b_4	$(4\mu - 3l + 2)/8$	$(4\mu - 3l - 2)/4$	P	odd $[1, (4\mu - 3l - 2)/4]$

Odd chain: b_4, b_3, b_0 . Even chain: b_1, b_2 .

FB6. $l \equiv 6 \pmod{8}$; domain $(3l + 10)/4 \leq \mu \leq l - 1$. Target order $(1, 3, 0, 4, 2)$. Lean: **famFB6**.

block	n_i	M_i	type	class runs
b_0	$(4\mu - l + 2)/8$	$(9l - 4\mu + 2)/4$	P	even $[(5l - 4\mu + 6)/4, l]$
b_1	$(l + 2)/8$	$(l - 2)/4$	P	odd $[1, (l - 2)/4]$
b_2	$(3l - 2)/8$	$(5l - 2)/4$	P	odd $[(l + 6)/4, l - 1]$
b_3	$l - \mu$	$-l/2$	N	even $[(4\mu - 3l + 6)/4, (5l - 4\mu - 2)/4]$
b_4	$(4\mu - 3l - 2)/8$	$(4\mu - 3l + 2)/4$	P	even $[2, (4\mu - 3l - 2)/4]$

Odd chain: b_1, b_2 . Even chain: b_4, b_3, b_0 .

F4. $l \equiv 6 \pmod{8}$, $\mu = (3l + 2)/4$; domain $l \geq 6$. Target order $(2, 0, 3, 1)$. Lean: **famF4**.

block	n_i	M_i	type	class runs
b_0	$(3l - 2)/8$	$(5l - 2)/4$	P	odd $[(l + 6)/4, l - 1]$
b_1	$(l + 2)/4$	$3l/2$	P	even $[(l + 2)/2, l]$
b_2	$(l - 2)/4$	$-l/2$	N	even $[2, (l - 2)/2]$
b_3	$(l + 2)/8$	$(l - 2)/4$	P	odd $[1, (l - 2)/4]$

Odd chain: b_3, b_0 . Even chain: b_2, b_1 .

A1. $l \equiv 0 \pmod{4}$, $\mu \equiv 1 \pmod{4}$; domain $l \geq 8$, $\mu \geq 5$, $l/2 + 1 \leq \mu \leq 3l/4 - 2$. Target order $(8, 3, 6, 7, 2, 0, 5, 1, 4)$. Lean: **famA1**.

block	n_i	M_i	type	class runs
b_0	$(\mu - 1)/4$	$(4l - \mu - 1)/2$	P	odd $[(2l - \mu + 3)/2, l - 1]$
b_1	1	$2l - 1$	P	even $\{l\}$
b_2	$(l - \mu + 1)/4$	$(3l - \mu - 5)/2$	P	even $[l/2, (2l - \mu - 3)/2]$
b_3	$(2\mu - l + 2)/4$	$(2\mu - l)/2$	P	odd $[1, (2\mu - l)/2]$
b_4	$(\mu - 1)/4$	$(4l - \mu - 5)/2$	P	even $[(2l - \mu + 1)/2, l - 2]$
b_5	$(\mu - 1)/4$	$(4l - 3\mu - 7)/2$	P	odd $[l - \mu, (2l - \mu - 5)/2]$
b_6	$(3l - 4\mu - 4)/4$	$(2 - l)/2$	N	odd $[(2\mu - l + 4)/2, l - \mu - 2]$
b_7	$(l - 4)/4$	$(2 - l)/2$	N	even $[2, (l - 4)/2]$
b_8	1	$\mu - 2l + 2$	N	odd $\{(2l - \mu - 1)/2\}$

Odd chain: b_3, b_6, b_5, b_8, b_0 . Even chain: b_7, b_2, b_4, b_1 .

A3. $l \equiv 0 \pmod{4}$, $\mu \equiv 3 \pmod{4}$; domain $l \geq 8$, $\mu \geq 7$, $l/2 + 1 \leq \mu \leq 3l/4 - 2$, $\mu \leq l - 5$. Target order $(8, 3, 6, 7, 2, 0, 5, 1, 4)$. Lean: **famA3**.

block	n_i	M_i	type	class runs
b_0	$(\mu + 1)/4$	$(4l - \mu - 3)/2$	P	odd $[(2l - \mu + 1)/2, l - 1]$
b_1	1	$2l - 1$	P	even $\{l\}$
b_2	$(l - \mu - 1)/4$	$(3l - \mu - 7)/2$	P	even $[l/2, (2l - \mu - 5)/2]$
b_3	$(2\mu - l + 2)/4$	$(2\mu - l)/2$	P	odd $[1, (2\mu - l)/2]$
b_4	$(\mu - 3)/4$	$(4l - \mu - 3)/2$	P	even $[(2l - \mu + 3)/2, l - 2]$
b_5	$(\mu + 1)/4$	$(4l - 3\mu - 5)/2$	P	odd $[l - \mu, (2l - \mu - 3)/2]$
b_6	$(3l - 4\mu - 4)/4$	$(2 - l)/2$	N	odd $[(2\mu - l + 4)/2, l - \mu - 2]$
b_7	$(l - 4)/4$	$(2 - l)/2$	N	even $[2, (l - 4)/2]$
b_8	1	$\mu - 2l + 2$	N	even $\{(2l - \mu - 1)/2\}$

Odd chain: b_3, b_6, b_5, b_0 . Even chain: b_7, b_2, b_8, b_4, b_1 .

U1. $l \equiv 0 \pmod{4}$, $\mu \equiv 1 \pmod{8}$; domain $3\mu \geq 2l + 11$, $5 \leq \mu \leq l - 3$, $\mu \leq 2l - 15$. Target order $(5, 7, 1, 3, 4, 0, 2, 6)$. Lean: **famU1**.

block	n_i	M_i	type	class runs
b_0	$(2l - \mu + 1)/8$	$(6l + \mu - 5)/4$	P	odd $[(2l + \mu + 3)/4, l - 1]$
b_1	$(3\mu - 2l - 3)/8$	$(2l + \mu + 3)/4$	P	even $[(2l - \mu + 9)/4, (\mu - 1)/2]$
b_2	$(\mu + 3)/4$	$(4l - \mu - 1)/2$	P	even $[(2l - \mu + 1)/2, l]$
b_3	$(2l - \mu + 1)/8$	$(2l - \mu - 3)/4$	P	odd $[1, (2l - \mu - 3)/4]$
b_4	$(2l - \mu - 7)/8$	$(2l - \mu - 3)/4$	P	even $[2, (2l - \mu - 7)/4]$
b_5	1	$(\mu - 2l + 1)/2$	N	even $\{(2l - \mu + 1)/4\}$
b_6	$(\mu - 1)/4$	$l - 1$	P	odd $[(2l - \mu + 5)/4, (2l + \mu - 5)/4]$
b_7	$(l - \mu - 1)/2$	$1 - l$	N	even $[(\mu + 3)/2, (2l - \mu - 3)/2]$

Odd chain: b_3, b_6, b_0 . Even chain: b_4, b_5, b_1, b_7, b_2 .

U3. $l \equiv 0 \pmod{4}$, $\mu \equiv 3 \pmod{8}$; domain $3\mu \geq 2l + 1$, $7 \leq \mu \leq l - 3$, $\mu \leq 2l - 29$. Target order $(5, 8, 0, 3, 6, 4, 1, 7, 2)$. Lean: **famU3**.

block	n_i	M_i	type	class runs
b_0	$(\mu - 3)/4$	$l + 1$	P	even $[(2l - \mu + 11)/4, (2l + \mu - 3)/4]$
b_1	$(\mu + 1)/4$	$(4l - \mu - 3)/2$	P	odd $[(2l - \mu + 1)/2, l - 1]$
b_2	$(2l - \mu + 3)/8$	$(6l + \mu + 1)/4$	P	even $[(2l + \mu + 5)/4, l]$
b_3	$(2l - \mu - 21)/8$	$(2l - \mu - 1)/4$	P	even $[4, (2l - \mu - 13)/4]$
b_4	1	$(2l - \mu + 1)/2$	P	even $\{(2l - \mu + 3)/4\}$
b_5	1	$(\mu - 2l + 7)/2$	N	even $\{(2l - \mu - 5)/4\}$
b_6	$(2l - \mu + 3)/8$	$(2l - \mu - 17)/4$	C1	odd $[1, (2l - \mu - 9)/4]$; even $\{2\}$
b_7	$(3\mu - 2l + 7)/8$	$(2l + \mu - 7)/4$	P	odd $[(2l - \mu - 1)/4, (\mu - 1)/2]$
b_8	$(l - \mu - 1)/2$	$1 - l$	N	odd $[(\mu + 3)/2, (2l - \mu - 3)/2]$

Odd chain: b_6^+, b_7, b_8, b_1 . Even chain: $b_6^-, b_3, b_5, b_4, b_0, b_2$.

U5. $l \equiv 0 \pmod{4}$, $\mu \equiv 5 \pmod{8}$; domain $3\mu \geq 2l + 7$, $5 \leq \mu \leq l - 3$, $\mu \leq 2l - 11$. Target order $(4, 7, 0, 3, 5, 1, 6, 2)$. Lean: **famU5**.

block	n_i	M_i	type	class runs
b_0	$(\mu - 1)/4$	$l + 1$	P	odd $[(2l - \mu + 9)/4, (2l + \mu - 1)/4]$
b_1	$(\mu + 3)/4$	$(4l - \mu - 1)/2$	P	even $[(2l - \mu + 1)/2, l]$
b_2	$(2l - \mu - 3)/8$	$(6l + \mu - 1)/4$	P	odd $[(2l + \mu + 7)/4, l - 1]$
b_3	$(2l - \mu - 3)/8$	$(2l - \mu + 1)/4$	P	even $[2, (2l - \mu - 3)/4]$
b_4	1	$(\mu - 2l + 1)/2$	N	odd $\{(2l - \mu + 1)/4\}$
b_5	$(2l - \mu - 3)/8$	$(2l - \mu - 7)/4$	P	odd $[1, (2l - \mu - 7)/4]$
b_6	$(3\mu - 2l + 1)/8$	$(2l + \mu - 1)/4$	P	even $[(2l - \mu + 5)/4, (\mu - 1)/2]$
b_7	$(l - \mu - 1)/2$	$1 - l$	N	even $[(\mu + 3)/2, (2l - \mu - 3)/2]$

Odd chain: b_5, b_4, b_0, b_2 . Even chain: b_3, b_6, b_7, b_1 .

U7. $l \equiv 0 \pmod{4}$, $\mu \equiv 7 \pmod{8}$; domain $5\mu \geq 4l + 11$, $7 \leq \mu \leq l - 1$, $\mu \leq 2l - 5$, $3\mu \leq 4l + 1$. Target order $(1, 6, 2, 4, 7, 0, 8, 3, 5)$. Lean: **famU7**.

block	n_i	M_i	type	class runs
b_0	$(\mu + 1)/8$	$(8l - \mu + 3)/4$	P	even $[(4l - \mu + 7)/4, l]$
b_1	$(\mu + 1)/8$	$(3\mu - 1)/4$	P	odd $[(\mu + 5)/4, (\mu - 1)/2]$
b_2	$(\mu + 1)/8$	$(8l - 5\mu - 1)/4$	P	even $[(4l - 3\mu + 5)/4, (2l - \mu - 1)/2]$
b_3	$(2l - \mu - 1)/4$	$(2l + \mu - 1)/2$	P	odd $[(\mu + 3)/2, l - 1]$
b_4	$(5\mu - 4l - 3)/8$	$(4l - 3\mu + 1)/4$	P	odd $[l - \mu + 2, (\mu - 3)/4]$
b_5	$(\mu + 1)/8$	$(8l - 3\mu + 1)/4$	P	even $[(2l - \mu + 3)/2, (4l - \mu - 1)/4]$
b_6	$l - \mu$	$(1 - \mu)/2$	N	even $[(5\mu - 4l + 5)/4, (4l - 3\mu - 3)/4]$
b_7	$(l - \mu + 1)/2$	$\mu - l$	N	odd $[1, l - \mu]$
b_8	$(5\mu - 4l - 3)/8$	$(5\mu - 4l + 1)/4$	P	even $[2, (5\mu - 4l - 3)/4]$

Odd chain: b_7, b_4, b_1, b_3 . Even chain: b_8, b_6, b_2, b_5, b_0 .

V. $l \equiv 0 \pmod{4}$, $\mu \equiv 3 \pmod{4}$; domain $5\mu \geq 3l + 11$, $3\mu \geq 2l + 9$, $2\mu \geq l + 6$, $7 \leq \mu \leq l - 5$, $5\mu \leq 4l + 3$. Target order $(8, 0, 5, 7, 3, 4, 1, 2, 6)$. Lean: **famV**.

block	n_i	M_i	type	class runs
b_0	$(l - \mu + 3)/4$	$(l + \mu - 1)/2$	P	even $[(\mu + 1)/2, l/2]$
b_1	$(\mu - 3)/4$	$(4l - \mu + 1)/2$	P	odd $[(2l - \mu + 5)/2, l - 1]$
b_2	$(\mu + 1)/4$	$(4l - \mu + 1)/2$	P	even $[(2l - \mu + 3)/2, l]$
b_3	$(4l - 5\mu + 7)/4$	$(2l - \mu + 1)/2$	P	odd $[(2\mu - l)/2, (3l - 3\mu + 3)/2]$
b_4	$(3\mu - 2l - 5)/4$	$(2l - \mu + 1)/2$	P	even $[l - \mu + 3, (\mu - 3)/2]$
b_5	$(5\mu - 3l - 7)/4$	$(\mu - l - 1)/2$	C3	even $[2, (3\mu - 2l - 5)/2]$; odd $[1, (2\mu - l - 4)/2]$
b_6	$(2\mu - l - 2)/4$	$(5l - 4\mu + 6)/2$	P	odd $[(3l - 3\mu + 7)/2, (2l - \mu + 1)/2]$
b_7	$(4l - 5\mu + 7)/4$	$(1 - \mu)/2$	N	even $[(3\mu - 2l - 1)/2, l - \mu + 1]$
b_8	$(l - \mu - 1)/4$	$(\mu - 3l - 1)/2$	N	even $[(l + 4)/2, (2l - \mu - 1)/2]$

Odd chain: b_5^-, b_3, b_6, b_1 . Even chain: $b_5^+, b_7, b_4, b_0, b_8, b_2$.

T. $l \equiv 0 \pmod{4}$, $\mu \equiv 3 \pmod{8}$; domain $9\mu \geq 8l + 11$, $11 \leq \mu \leq l - 1$, $3\mu \leq 4l - 3$, $7\mu \leq 8l - 3$. Target order $(2, 3, 6, 0, 4, 1, 5)$. Lean: **famT**.

block	n_i	M_i	type	class runs
b_0	$(\mu + 1)/4$	$(4l - \mu + 1)/2$	P	even $[(2l - \mu + 3)/2, l]$
b_1	$(\mu - 3)/8$	$(8l - \mu - 1)/4$	P	odd $[(4l - \mu + 7)/4, l - 1]$
b_2	$(8l - 7\mu + 5)/8$	$(\mu + 1)/4$	C3	even $[2, (4l - 3\mu + 1)/4]$; odd $[1, l - \mu]$
b_3	$(9\mu - 8l - 3)/8$	$(\mu + 1)/4$	P	odd $[l - \mu + 2, (5\mu - 4l - 3)/4]$
b_4	$(\mu - 3)/8$	$(8l - 5\mu + 3)/4$	P	even $[(4l - 3\mu + 9)/4, (2l - \mu - 1)/2]$
b_5	$(\mu + 1)/4$	$2l - \mu$	P	odd $[(4l - 3\mu + 5)/4, (4l - \mu - 1)/4]$
b_6	$l - \mu$	$(1 - \mu)/2$	N	odd $[(5\mu - 4l + 5)/4, (4l - 3\mu - 3)/4]$

Odd chain: $b_2^-, b_3, b_6, b_5, b_1$. Even chain: b_2^+, b_4, b_0 .

FA3. $l \equiv 1 \pmod{4}$, $\mu \equiv 3 \pmod{4}$; domain $(l+5)/2 \leq \mu \leq l-6$. Target order $(6, 1, 3, 4, 0, 5, 2)$. Lean: **famFA3**.

block	n_i	M_i	type	class runs
b_0	$(\mu+1)/4$	$(4l-\mu+1)/2$	P	odd $[(2l-\mu+3)/2, l]$
b_1	$(\mu+1)/4$	$(2l-\mu-1)/2$	P	even $[(l-\mu+2)/2, (l-1)/2]$
b_2	$(l-1)/4$	$(3l-1)/2$	P	even $[(l+3)/2, l-1]$
b_3	$(l-\mu-2)/4$	$(l-\mu)/2$	P	even $[2, (l-\mu-2)/2]$
b_4	$(l-\mu+2)/4$	$(l-\mu)/2$	P	odd $[1, (l-\mu)/2]$
b_5	$(2\mu-l-1)/4$	$(l+1)/2$	P	odd $[(l-\mu+4)/2, (\mu-1)/2]$
b_6	$(l-\mu)/2$	$-l$	N	odd $[(\mu+3)/2, (2l-\mu-1)/2]$

Odd chain: b_4, b_5, b_6, b_0 . Even chain: b_3, b_1, b_2 .

FA1. $l \equiv 1 \pmod{4}$, $\mu \equiv 1 \pmod{4}$; domain $(l+5)/2 \leq \mu \leq l-4$. Target order $(7, 1, 3, 4, 0, 5, 2, 6)$. Lean: **famFA1**.

block	n_i	M_i	type	class runs
b_0	$(\mu-1)/4$	$(4l-\mu-1)/2$	P	even $[(2l-\mu+3)/2, l-1]$
b_1	$(\mu-5)/4$	$(2l-\mu+1)/2$	P	odd $[(l-\mu+6)/2, (l-3)/2]$
b_2	$(l-1)/4$	$(3l+3)/2$	P	odd $[(l+5)/2, l]$
b_3	$(l-\mu)/4$	$(l-\mu+2)/2$	P	even $[2, (l-\mu)/2]$
b_4	$(l-\mu+4)/4$	$(l-\mu+2)/2$	P	odd $[1, (l-\mu+2)/2]$
b_5	$(2\mu-l-1)/4$	$(l+1)/2$	P	even $[(l-\mu+4)/2, (\mu-1)/2]$
b_6	1	l	P	odd $\{(l+1)/2\}$
b_7	$(l-\mu)/2$	$-l$	N	even $[(\mu+3)/2, (2l-\mu-1)/2]$

Odd chain: b_4, b_1, b_6, b_2 . Even chain: b_3, b_5, b_7, b_0 .

FT. $l \equiv 1 \pmod{4}$, $\mu = l-2$; domain $l \geq 9$. Target order $(5, 1, 3, 0, 4, 2)$. Lean: **famFT**.

block	n_i	M_i	type	class runs
b_0	$(l-1)/4$	$(3l-1)/2$	P	even $[(l+3)/2, l-1]$
b_1	$(l-5)/4$	$(l+1)/2$	P	odd $[3, (l-3)/2]$
b_2	$(l-1)/4$	$(3l+3)/2$	P	odd $[(l+5)/2, l]$
b_3	1	1	P	odd $\{1\}$
b_4	$(l-1)/4$	$(l+1)/2$	P	even $[2, (l-1)/2]$
b_5	1	$-l$	N	odd $\{(l+1)/2\}$

Odd chain: b_3, b_1, b_5, b_2 . Even chain: b_4, b_0 .

FL1. $l \equiv 1 \pmod{8}$, $\mu = (l+1)/2$; domain $l \geq 17$. Target order $(6, 0, 2, 4, 5, 1, 3)$. Lean: **famFL1**.

block	n_i	M_i	type	class runs
b_0	1	l	P	odd $\{(l+1)/2\}$
b_1	$(l-1)/8$	$(7l-3)/4$	P	even $[(3l+5)/4, l-1]$
b_2	$(l-9)/8$	$(3l+1)/4$	P	odd $[(l+11)/4, (l-3)/2]$
b_3	$(l-1)/4$	$(3l+3)/2$	P	odd $[(l+5)/2, l]$
b_4	$(l-1)/8$	$(l+3)/4$	P	even $[2, (l-1)/4]$
b_5	$(l+7)/8$	$(l+3)/4$	P	odd $[1, (l+3)/4]$
b_6	$(l-1)/4$	$-l$	N	even $[(l+7)/4, (3l-3)/4]$

Odd chain: b_5, b_2, b_0, b_3 . Even chain: b_4, b_6, b_1 .

FL5. $l \equiv 5 \pmod{8}$, $\mu = (l+1)/2$; domain $l \geq 13$. Target order $(5, 1, 3, 4, 0, 2)$. Lean: **famFL5**.

block	n_i	M_i	type	class runs
b_0	$(l+3)/8$	$(7l+1)/4$	P	odd $[(3l+5)/4, l]$
b_1	$(l+3)/8$	$(3l-3)/4$	P	even $[(l+3)/4, (l-1)/2]$
b_2	$(l-1)/4$	$(3l-1)/2$	P	even $[(l+3)/2, l-1]$
b_3	$(l-5)/8$	$(l-1)/4$	P	even $[2, (l-5)/4]$
b_4	$(l+3)/8$	$(l-1)/4$	P	odd $[1, (l-1)/4]$
b_5	$(l-1)/4$	$-l$	N	odd $[(l+7)/4, (3l-3)/4]$

Odd chain: b_4, b_5, b_0 . Even chain: b_3, b_1, b_2 .

R30. $l \equiv 3 \pmod{4}$, $\mu \equiv 1 \pmod{4}$; domain $(l+3)/2 \leq \mu \leq l-6$. Target order $(6, 1, 3, 4, 0, 5, 2)$. Lean: **famR30**.

block	n_i	M_i	type	class runs
b_0	$(\mu-1)/4$	$(4l-\mu-1)/2$	P	even $[(2l-\mu+3)/2, l-1]$
b_1	$(\mu-1)/4$	$(2l-\mu+1)/2$	P	odd $[(l-\mu+4)/2, (l-1)/2]$
b_2	$(l+1)/4$	$(3l+1)/2$	P	odd $[(l+3)/2, l]$
b_3	$(l-\mu-2)/4$	$(l-\mu)/2$	P	even $[2, (l-\mu-2)/2]$
b_4	$(l-\mu+2)/4$	$(l-\mu)/2$	P	odd $[1, (l-\mu)/2]$
b_5	$(2\mu-l+1)/4$	$(l-1)/2$	P	even $[(l-\mu+2)/2, (\mu-1)/2]$
b_6	$(l-\mu)/2$	$-l$	N	even $[(\mu+3)/2, (2l-\mu-1)/2]$

Odd chain: b_4, b_1, b_2 . Even chain: b_3, b_5, b_6, b_0 .

R3E. $l \equiv 3 \pmod{4}$, $\mu \equiv 3 \pmod{4}$; domain $(l+7)/2 \leq \mu \leq l-4$. Target order $(7, 0, 2, 4, 5, 1, 6, 3)$. Lean: **famR3E**.

block	n_i	M_i	type	class runs
b_0	1	l	P	even $\{(l+1)/2\}$
b_1	$(\mu+1)/4$	$(4l-\mu+1)/2$	P	odd $[(2l-\mu+3)/2, l]$
b_2	$(\mu-3)/4$	$(2l-\mu-1)/2$	P	even $[(l-\mu+4)/2, (l-3)/2]$
b_3	$(l-3)/4$	$(3l+1)/2$	P	even $[(l+5)/2, l-1]$
b_4	$(l-\mu)/4$	$(l-\mu+2)/2$	P	even $[2, (l-\mu)/2]$
b_5	$(l-\mu+4)/4$	$(l-\mu+2)/2$	P	odd $[1, (l-\mu+2)/2]$
b_6	$(2\mu-l-3)/4$	$(l+3)/2$	P	odd $[(l-\mu+6)/2, (\mu-1)/2]$
b_7	$(l-\mu)/2$	$-l$	N	odd $[(\mu+3)/2, (2l-\mu-1)/2]$

Odd chain: b_5, b_6, b_7, b_1 . Even chain: b_4, b_2, b_0, b_3 .

R3Q. $l \equiv 3 \pmod{4}$, $\mu \equiv 3 \pmod{4}$; domain $(l+3)/2 \leq \mu \leq (2l-1)/3$. Target order $(6, 0, 2, 4, 5, 1, 3)$. Lean: **famR3Q**.

block	n_i	M_i	type	class runs
b_0	$(2\mu-l-1)/2$	l	P	even $[l-\mu+2, \mu-1]$
b_1	$(\mu+1)/4$	$(4l-\mu+1)/2$	P	odd $[(2l-\mu+3)/2, l]$
b_2	$(2l-3\mu+3)/4$	$(2l-\mu-1)/2$	P	even $[(\mu+1)/2, l-\mu]$
b_3	$(l-\mu)/2$	$l+\mu-1$	P	even $[\mu+1, l-1]$
b_4	$(\mu-3)/4$	$(\mu-1)/2$	P	even $[2, (\mu-3)/2]$
b_5	$(\mu+1)/4$	$(\mu-1)/2$	P	odd $[1, (\mu-1)/2]$
b_6	$(l-\mu)/2$	$-l$	N	odd $[(\mu+3)/2, (2l-\mu-1)/2]$

Odd chain: b_5, b_6, b_1 . Even chain: b_4, b_2, b_0, b_3 .

R3T1. $l \equiv 3 \pmod{4}$, $\mu = l - 2$; domain $l \geq 7$. Target order $(5, 1, 3, 0, 4, 2)$. Lean: `famR3T1`.

block	n_i	M_i	type	class runs
b_0	$(l-3)/4$	$(3l+1)/2$	P	even $[(l+5)/2, l-1]$
b_1	$(l-3)/4$	$(l+3)/2$	P	odd $[3, (l-1)/2]$
b_2	$(l+1)/4$	$(3l+1)/2$	P	odd $[(l+3)/2, l]$
b_3	1	1	P	odd $\{1\}$
b_4	$(l-3)/4$	$(l-1)/2$	P	even $[2, (l-3)/2]$
b_5	1	$-l$	N	even $\{(l+1)/2\}$

Odd chain: b_3, b_1, b_2 . Even chain: b_4, b_5, b_0 .

TT. l odd, $\mu = l$; domain $l \geq 3$ (this is τ_l). Target order $(0, 1)$. Lean: `tauGraph`.

block	n_i	M_i	type	class runs
b_0	$(l+1)/2$	l	P	odd $[1, l]$
b_1	$(l-1)/2$	l	P	even $[2, l-1]$

Odd chain: b_0 . Even chain: b_1 .

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